Zero dimension for the real exceptional set and vanishing 3-adic nesting constants

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Abstract

Let $\Sigma \subset \mathbb{Z}_3$ be the ternary Cantor set with digits $0, 1$. We determine the generalized exceptional-set dimension and nesting constant exactly: $\dim_{\mathrm{H}} E_* = \dim_{\mathrm{P}} E_* = \Gamma_* = (\log_3 2)/2$. This matches the lower bound of Abram–Lagarias, based on Bolshakov's construction. The proof supplies a quantitative finite-intersection bound and a sharp Assouad bound for each fixed infinite family of multipliers. Independently, for Haar-almost every $\lambda \in 1 + 3\mathbb{Z}_3$, the Hausdorff, packing and lower and upper box dimensions of $\Sigma \cap \lambda^{-1}\Sigma$ all equal $\log_3(4/3)$. We prove this through an exact second moment, positivity of a convolution density and a block-measure transfer to the homogeneous intersection. An isometric coding gives the sharp golden-ratio bound for arbitrary nonidentity 3-adic unit multipliers. A further affine transfer proves zero Hausdorff and packing dimension for the circle exceptional set and for the stated fixed moving targets.

These results are added in version 4. The manuscript retains the complete proofs from versions 1–3: zero Hausdorff and packing dimension for the 3-adic and full real exceptional sets of powers of two; vanishing dyadic finite-visit nesting constants; generic absence of three nonzero visits; exact digital rigidity; and linear-order counting of positive-dimensional integer multipliers, including a linear gap from finite Cantor quotients. The prescribed-point assertion $1 \notin E_2$ remains unresolved.

Version history and stable citation. This is version 4.0.0 of the same manuscript, with the same author, title and citation key `iuzhakov2026lagarias`. Version 1.0.0 proved zero Hausdorff and packing dimension for E_2. Version 2.0.0 added uniform finite-visit covers, vanishing nesting constants and the full real exceptional-set theorem. Version 3.0.0 added the metric and linear-counting results in Sections 13 and 14. Their proof sections and the causal-container appendix are retained. The new proofs occupy Sections 15, 16 and 17. Earlier versioned releases remain unchanged; citations should identify the version and theorem used.

What is new in version 4. Section 15 determines E_* and Γ_* exactly, with a finite-visit convergence estimate and a fixed-family Assouad strengthening. This is a two-window causal argument, with its complete palette and entropy proof given here. Section 16 determines the Haar-typical intersection dimension, using an independent analytic argument. It also extends the integer-parameter golden-ratio coding of [2] to arbitrary 3-adic units and describes the punctured dimension oscillation at 1. Section 17 transfers the previously proved uniform affine estimates to fixed moving targets, including the circle problem. These are separate proof branches, not steps of a proof for the point 1.

Which questions are answered. The retained zero-dimensional assertions concern Lagarias's Conjectures A and B in [5], numbered 1.4 and 1.7 in the journal version [1]. Version

3 answered the order-of-growth part of his Problem 2 (journal Problem 1.11), without determining a density constant. The new exact generalized dimension sharpens the bounds in Abram–Lagarias's generalized exceptional-set Problem 1.6 and identifies their nesting constant [7]. Its positive lower bound was already known; the new contribution is the matching upper bound and strengthenings. The typical-value theorem answers the Haar-generic part of the parameter programme in [7, Section 2.4], not the full parameter spectrum or the Baire-category problem. The circle corollary answers the separate question in [5, Section 5]. The finite-quotient results have their own manuscript [9]. No exhaustive external-priority claim is inferred from internal review.

The prescribed point remains unresolved. The generalized constant Γ_* is different from the dyadic constant $\Gamma_{\mathrm{H}} = 0$ retained from version 2. A Haar-almost-everywhere result does not decide a prescribed rational multiplier. In particular, for $b = 2$ and starting point 1 the known visits $1, 4, 256$ occur at times $0, 2, 8$; this base is exceptional for the generic three-visit theorem. Neither the dimension theorems nor the circle corollary proves that these are the only visits.

1 The 3-adic finite-visit problem

We use the metric $|x - y|_3$ on $\mathbb{Z}_3$, with $v_3(3) = 1$ and $v_3(0) = +\infty$. Put

$$\Sigma = \left\{ \sum_{j \geq 0} x_j 3^j : x_j \in \{0, 1\} \right\}, \qquad F_k = \{x \in \mathbb{Z}_3 : \#\{n \geq 0 : 2^n x \in \Sigma\} \geq k\}.$$

The visits in this definition occur at distinct nonnegative integer times. Set

$$\Gamma_{\mathrm{H}} = \lim_{k \to \infty} \dim_{\mathrm{H}} F_k, \qquad \Gamma_{\mathrm{P}} = \lim_{k \to \infty} \dim_{\mathrm{P}} F_k.$$

These limits exist because the sets decrease. The Hausdorff limit is the dyadic nesting constant considered in the study of the exceptional set by Lagarias and by Abram, Bolshakov and Lagarias [1, 2]. We retain the arbitrary-visit convention throughout: in particular $F_1 = \bigcup_{n \geq 0} 2^{-n}\Sigma$, rather than literally Σ.

Theorem 1.1. *For every $\delta > 0$ there is an integer $J \geq 1$ and a countable family of compact subsets $K_m \subset \mathbb{Z}_3$ such that*

$$F_J \subseteq \bigcup_{m \geq 1} K_m, \qquad \overline{\dim}_{\mathrm{B}} K_m \leq \delta \quad (m \geq 1).$$

Consequently

$$\boxed{\Gamma_{\mathrm{H}} = \Gamma_{\mathrm{P}} = 0.}$$

The distinction from an infinite-visit argument is the order of quantifiers. The cardinality J will be fixed before any visiting times are given. The exponent spans of the extracted templates need not be bounded uniformly. They are fixed separately on each member of the countable cover before its digit length tends to infinity.

The analytic input is the finite causal-container theorem in Appendix A of the frozen release v1.0.0 of [4]. The real theorem is stated in Theorem 7.1; its proof begins in Section 7 and uses a separate arithmetic extraction. We state exactly the portion used in Section 5 and reproduce its complete source unchanged in Appendix A. The new arithmetic extraction is proved in full below. In the 3-adic finite-visit argument, neither the extraction from an infinite set in [4] nor a uniform bound on the spans of finite templates is used.

2 Uniform finite extraction: statement and incidence bound

For a finite set of nonnegative integers P write

$$\Delta(P) = \{v - u : u, v \in P,\ u < v\}.$$

These are sets, so equal distances within one P are counted only once.

Theorem 2.1 (Uniform finite templates)**.** *Fix integers $r \geq 3$, $K \geq 1$, and $q \geq 1$. There is a finite integer $J = J(r, K, q)$ with the following property. Every set of J distinct powers 2^n, $n \geq 0$, contains K distinct members γ_i and admits rational units*

$$\alpha, A_1, \ldots, A_{r-2}, u_1, \ldots, u_K \in \mathbb{Q} \cap \mathbb{Z}_3^\times$$

such that

$$c_i := \gamma_i/\alpha = 1 + \sum_{t=1}^{r-2} A_t 3^{e_{ti}} + u_i 3^{E_i} \qquad (1 \leq i \leq K). \tag{1}$$

The positions

$$P_i = \{0, e_{1i}, \ldots, e_{r-2,i}, E_i\}$$

are in increasing order, every successive gap is at least q, and $\Delta(P_i) \cap \Delta(P_j) = \varnothing$ for $i \neq j$. The bound J is independent of the original powers and of all exponent spans. The coefficient α is a virtual unit anchor: it need not be one of the original powers.

We prove this theorem by induction on the number of common monomial coordinates. At a stage with $d \geq 1$ coordinates the data are

$$\gamma_i = 2^{n_i} \quad \text{distinct}, \qquad 0 = e_{0i} < e_{1i} < \cdots < e_{d-1,i}, \tag{2}$$

$$\gamma_i = \sum_{t=0}^{d-1} a_t 3^{e_{ti}} + \varepsilon_i, \quad a_t \in \mathbb{Q} \cap \mathbb{Z}_3^\times, \quad v_3(\varepsilon_i) \geq e_{d-1,i} + q. \tag{3}$$

Successive gaps in (2) are at least q, and every column $e_{ti} - e_{si}$, $s < t$, is injective as i varies. For $d = 1$ the last condition is empty.

We use the uniform theorem of Evertse, Schlickewei and Schmidt [3, Theorem 1.1]. In characteristic zero, an equation $a_1 z_1 + \cdots + a_h z_h = 1$, with nonzero coefficients and variables in a subgroup of $(L^\times)^h$ of rank ρ, has at most

$$\exp\big((6h)^{3h}(\rho + 1)\big)$$

nondegenerate solutions. Here nondegenerate means that no nonempty subsum on the left vanishes. The theorem applies over an algebraic closure of $\mathbb{Q}_3$ and hence to the solutions considered here.

Lemma 2.2 (Uniform incidence bound)**.** *Suppose the γ_i are distinct powers of two and every difference column $e_{ti} - e_{si}$ is injective. For any fixed coefficients $c_0, \ldots, c_{d-1} \in \mathbb{Q}_3$, the equation*

$$\gamma_i = \sum_{t=0}^{d-1} c_t 3^{e_{ti}} \tag{4}$$

holds for at most

$$B_d = \binom{d}{2} + \left\lceil \exp\big((6d)^{3d}(d + 2)\big) \right\rceil$$

indices i.

Proof. Delete zero coefficients. If no coefficient remains, there is no solution. For each pair of remaining terms, equality of valuations fixes one value of $e_{ti} - e_{si}$ and occurs on at most one row. Excluding at most $\binom{d}{2}$ rows, all terms on the right of (4) have different valuations; no nonempty subsum of them can vanish.

Choose one remaining index t_0 and divide by $c_{t_0} 3^{e_{t_0 i}}$:

$$c_{t_0}^{-1} \gamma_i 3^{-e_{t_0 i}} - \sum_{t \neq t_0} \frac{c_t}{c_{t_0}} 3^{e_{ti} - e_{t_0 i}} = 1. \tag{5}$$

A vanishing subsum not containing the γ_i term is impossible by the distinct valuations. A vanishing subsum containing that term would, on using (4), force a nonempty complementary subsum of the original right-hand side to vanish; the complement includes the t_0 term. Thus (5) is nondegenerate.

Its variable tuple belongs to the image of $\langle 2 \rangle \times \langle 3 \rangle^d$, a group of rank at most $d+1$. Two rows with the same tuple have $2^{n_i - n_j} = 3^{e_{t_0 i} - e_{t_0 j}}$, hence $n_i = n_j$ by multiplicative independence. Thus rows give distinct solutions. The uniform unit-equation bound gives the stated number. If only one coefficient remains, multiplicative independence gives at most one row directly. □

3 The extension lemma

Lemma 3.1 (One additional coordinate)**.** *For every $d \geq 1$ and $L \geq 1$ there is a finite $N_d(L)$ such that, from $N_d(L)$ rows satisfying (2)–(3) and the injectivity conditions above, one can retain L rows and find rational units $b_0, \ldots, b_{d-1}$ with*

$$\gamma_i = \sum_{t=0}^{d-1} b_t 3^{e_{ti}} + u_i 3^{E_i}, \quad u_i \in \mathbb{Q} \cap \mathbb{Z}_3^{\times}, \quad E_i \geq e_{d-1,i} + q, \tag{6}$$

such that every column $E_i - e_{ti}$, $0 \leq t < d$, is injective. The previous exponent columns do not change. The threshold $N_d(L)$ can be chosen independently of q, the coefficients and the exponents.

3.1 Growth order and valuation-minimal minors

Give the rows any linear order. Color a pair $i < j$ by the signs of all differences

$$(e_{tj} - e_{sj}) - (e_{ti} - e_{si}) \qquad (s < t).$$

None is zero. Finite Ramsey theory gives an arbitrarily large homogeneous subset, provided the original set is sufficiently large. The numbers $\Delta_t(i,j) = e_{tj} - e_{ti}$ have the same strict order for every pair $i < j$: their pairwise comparisons are fixed, and they are transitive since they compare actual numbers. Temporarily relabel the coordinates $1, \ldots, d$ in this growth order, so that

$$\Delta_1(i,j) < \cdots < \Delta_d(i,j) \qquad (i < j). \tag{7}$$

This temporary order need not agree with the increasing order of exponents within a row.

For $s < t$ and $i < j$, (7) gives

$$e_{si} + e_{tj} > e_{sj} + e_{ti}. \tag{8}$$

It follows that every square submatrix of $(3^{e_{ti}})$ has exactly one determinant term of least valuation. Indeed, exchanging an assignment which is not order-reversing strictly lowers its

valuation by (8). The unique minimum pairs increasing row indices with decreasing growth-coordinate indices. In particular every d-by-d coordinate minor is nonzero, and its valuation equals that minimum permutation cost.

Adjoin γ_i to the coordinate column and write

$$v_i = (3^{e_{1i}}, \ldots, 3^{e_{di}}, \gamma_i)^{\mathsf{T}} \in \mathbb{Q}_3^{d+1}.$$

Color each $(d+1)$-subset according to whether its augmented determinant is zero. An all-zero subset of size greater than B_d is impossible: its augmented matrix would have rank at most d, while the coordinate matrix has rank d, forcing the γ_i row to be one fixed linear combination of the coordinate rows, contrary to Lemma 2.2. A second Ramsey refinement therefore gives as many rows as needed with every augmented $(d+1)$-minor nonzero.

3.2 Canonical determinant valuations

For an increasing $(d+1)$-tuple $I = (i_1, \ldots, i_{d+1})$ put

$$D_I = \det(v_{i_1}, \ldots, v_{i_{d+1}}), \qquad w_I = v_3(D_I).$$

Consider an increasing $(d+2)$-tuple. For each slot $h \in \{1, \ldots, d+1\}$, compare its two $(d+1)$-subtuples obtained by keeping either of the adjacent indices in positions $h, h+1$ and keeping every other index. Color this comparison $t \in \{1, \ldots, d\}$ if the change in w equals the change in e_t at the variable slot; otherwise color it G. At most one coordinate color is possible by (7). There are at most $(d+1)^{d+1}$ combined colors. Another finite Ramsey refinement makes the comparison color at each slot independent of the chosen indices.

If one slot has color G, fix the d other indices as pivots, with a large consecutive group of variable indices in that slot. For these variables, each column

$$w_{\text{pivots},i} - e_{ti} \qquad (1 \le t \le d) \tag{9}$$

is injective. This is the situation needed for interpolation below. We now prove that a G slot must exist on a sufficiently large homogeneous subset.

Suppose every slot has a coordinate color t_h. The graph of increasing $(d+1)$-tuples, with an edge for a change in only one coordinate, is connected. Along such an edge the value $w_I - \sum_h e_{t_h,i_h}$ is unchanged, because the union of the two tuples is an increasing $(d+2)$-tuple of the type just colored. Consequently there is one constant C with

$$w_{i_1,\ldots,i_{d+1}} = C + \sum_{h=1}^{d+1} e_{t_h,i_h} \tag{10}$$

for every increasing tuple in the homogeneous subset.

For a coordinate t and an increasing $(d+2)$-tuple I, the cofactor identity is

$$\sum_{l=1}^{d+2} (-1)^l 3^{e_{t,i_l}} D_{I \setminus i_l} = 0. \tag{11}$$

Its nonzero summands cannot have a unique least valuation. Color I by a selected pair (l_t, m_t), $l_t < m_t$, of minimum-valuation positions for each t. A further Ramsey refinement fixes these pairs for all tuples. This uses at most $\binom{d+2}{2}^d$ colors.

Write A_l for the valuation of the lth summand of (11). For a fixed t and its fixed pair $l < m$, the identity $A_m - A_l = 0$, after inserting (10), has the following slot contributions:

$$\begin{cases} e_{t_l,i_l} - e_{t,i_l}, & \text{at slot } l, \\ e_{t_j,i_j} - e_{t_{j-1},i_j}, & l < j < m, \\ e_{t,i_m} - e_{t_{m-1},i_m}, & \text{at slot } m. \end{cases} \tag{12}$$

All other contributions cancel. Hold the other slots fixed and give one slot two possible indices. This is possible for every slot in an ordered set with at least $d+3$ elements. A difference of two distinct coordinate functions is injective by hypothesis, so it cannot have the same value for these two choices. Hence all the labels in each slot function of (12) coincide, and

$$t_l = t_{l+1} = \cdots = t_{m-1} = t. \tag{13}$$

In particular each of the d coordinate labels appears in the word $t_1, \ldots, t_{d+1}$. Exactly one label appears twice and all the others appear once.

This word is nonincreasing in the growth order. Indeed, adjacent summand valuations for coordinate t satisfy

$$A_{j+1} - A_j = \Delta_t(i_j, i_{j+1}) - \Delta_{t_j}(i_j, i_{j+1}). \tag{14}$$

Suppose $t_h = a < b = t_{h+1}$. If a appeared only once, its fixed minimum pair, by (13), would be $h, h+1$. But the next difference in (14) is negative, making A_{h+2} smaller than that minimum. Thus a must be the repeated label. If b appeared only once, its minimum pair would be $h+1, h+2$, while the preceding difference in (14) is positive, making A_h smaller than that minimum. Thus b must also repeat, a contradiction. Therefore the word has the form

$$d, d-1, \ldots, t_*, t_*, \ldots, 1. \tag{15}$$

3.3 A two-dimensional residue-field obstruction

Let the repeated label t_* occupy slots $h, h+1$. Fix $h-1$ pivot indices before a group of five variable indices, and $d-h$ pivots after it. There are $d-1$ pivots in total, so $d+4$ available indices suffice. Their augmented columns are independent, as follows by extending them to a nonzero d-by-d coordinate minor. Quotient $\mathbb{Q}_3^{d+1}$ by their span. This is a two-dimensional vector space; let u_i denote the images of the five variable columns.

In any fixed coordinates on the quotient, its pair determinants are the corresponding augmented determinants times one fixed nonzero factor (up to signs). By (10) and (15),

$$v_3 \det(u_i, u_j) = C' + e_{t_*,i} + e_{t_*,j}.$$

The vectors $z_i = 3^{-e_{t_*,i}} u_i$ therefore have all pair determinants nonzero and of the same valuation. There can be at most four such vectors. To see this without any normalization assumption on their norms, use z_1, z_2 as a basis and write $z_i = a_i z_1 + b_i z_2$ for $i \geq 3$. Comparison with the first two vectors gives $a_i, b_i \in \mathbb{Z}_3^\times$. Comparison between $i, j \geq 3$ gives $a_i b_j - a_j b_i \in \mathbb{Z}_3^\times$. Thus the residues of a_i/b_i are distinct nonzero elements of $\mathbb{F}_3$, of which there are only two. There are at most two vectors beyond the basis, contradicting the five variables. The assumption that all slots have coordinate colors is impossible. A G slot exists.

3.4 Interpolation and preservation of the gap

Restore the original within-row exponent order $0 = e_{0i} < \cdots < e_{d-1,i}$. Fix the d pivots of a G slot. Their coordinate matrix is invertible. Let b_t interpolate the γ_i on the pivots, and put $\delta_t = b_t - a_t$. The data for δ_t are the errors ε_i.

By Cramer's rule, $v_3(\delta_t) \geq q$ for every t. In fact, the denominator determinant has valuation exactly its minimum permutation cost, by (8). A term of a numerator determinant replaces one entry $3^{e_{ti}}$ in the corresponding denominator term by ε_i, whose valuation is at least $\max_s e_{si} + q \geq e_{ti} + q$. Every numerator term has valuation at least the denominator's minimum cost plus q. Cancellation can only increase this valuation. The b_t remain units; they are rational, since both the coordinate matrix and its interpolation data are rational.

For a variable index put

$$R_i = \gamma_i - \sum_{t=0}^{d-1} b_t 3^{e_{ti}}, \qquad E_i = v_3(R_i).$$

The augmented determinant formula for an interpolation residual gives $E_i = w_{\mathrm{pivots},i} - C_0$ for one fixed constant C_0. All residuals are nonzero, and (9) shows that every $E_i - e_{ti}$ is injective.

It remains to discard rows on which $E_i < e_{d-1,i} + q$. On such a row the old ε_i cannot attain the least valuation in

$$R_i = \varepsilon_i - \sum_t \delta_t 3^{e_{ti}}.$$

If two *nonzero* error terms tie for the finite least valuation, that equality fixes one value of the corresponding difference column; there is at most one such row per unordered pair of coordinates. These rows number at most $\binom{d}{2}$. On every other bad row exactly one error term, say t, is least, giving

$$E_i - e_{ti} = v_3(\delta_t).$$

Injectivity of this residual column permits at most one such row for each t, hence at most d more. Deleting at most $d + \binom{d}{2}$ rows gives (6) and proves Lemma 3.1, subject only to choosing the finite cardinality thresholds, made explicit next.

3.5 Uniform cardinality bookkeeping

Let $\mathcal{R}_k(m; c)$ be any finite Ramsey number for c colors on k-subsets and monochromatic size m; use the usual one-color convention. For desired output L, put

$$\begin{aligned}
b_d &= d + \binom{d}{2}, \qquad M = L + b_d, \\
H &= \max\left\{ d + M,\ \mathcal{R}_{d+2}\left(d+4; \binom{d+2}{2}^{d} \right) \right\}, \\
P &= \mathcal{R}_{d+2}\left(H; (d+1)^{d+1} \right), \\
Q_0 &= \mathcal{R}_{d+1}\left(\max\{P, B_d + 1\}; 2 \right), \\
N_d(L) &= \mathcal{R}_2\left(Q_0; 2^{\binom{d}{2}} \right).
\end{aligned}$$

The refinements are respectively the growth comparisons, the nonzero augmented minors, and the fixed slot colors. A G slot leaves at least $H - d \geq L + b_d$ variable indices: take the required pivots at the two ends and all intermediate indices as variables. Discarding b_d bad rows leaves L. If there is no G slot, the minimum-pair refinement leaves $d + 4$ indices, sufficient for every local slot variation and the five-variable quotient contradiction. This supplies a finite threshold depending only on d, L. In particular, no valuation depth or exponent span occurs in it.

4 Iteration and disjoint distance sets

Proof of Theorem 2.1. There are $U_q = 2 \cdot 3^{q-1}$ unit residue classes modulo 3^q. Pigeonhole sufficiently many original powers into one such class and choose an ordinary unit representative a_0. Then $\gamma_i = a_0 + \varepsilon_i$, with $v_3(\varepsilon_i) \geq q$, provides the $d = 1$ starting stage.

Apply Lemma 3.1. If another common coordinate is needed, write its residual $R_i = u_i 3^{E_i}$ and retain one residue class of the units u_i modulo 3^q, represented by a unit a_d. Adjoin the common monomial $a_d 3^{e_{di}}$ with $e_{di} = E_i$. The remaining error has valuation at least $e_{di} + q$. All former difference columns are unchanged, and all new ones are injective by the extension lemma. Thus the induction hypotheses hold with $d + 1$ common coordinates.

At $d = r - 1$ perform one final extension, leaving its terminal unit coefficient free. There are then r positions, every successive gap is at least q, and each of the $c = \binom{r}{2}$ positive difference columns is injective. Form the graph on rows in which two rows are adjacent if their distance sets intersect. A fixed row has at most c distances. For each of these distances, each of the c difference columns attains it in at most one row. Thus the graph has maximum degree at most c^2. From $K(c^2 + 1)$ rows a greedy independent-set selection retains K pairwise disjoint distance sets. This argument allows repeated distances within an individual template.

For clarity, all input sizes may be fixed backwards. Start with $L_{r-1} = K(c^2 + 1)$ and require $N_{r-1}(L_{r-1})$ rows at that stage. For each preceding extension choose an output size U_q times the next stage's required input size, and apply the corresponding N_d. Finally multiply the $d = 1$ required input size by U_q for the initial residue pigeonhole. This is a finite integer $J(r, K, q)$ depending only on the displayed parameters.

The final common coefficient at exponent zero is a rational unit α. Divide the identity by α to obtain (1). All coefficients and terminal residuals are rational, because every interpolation used rational data. No bound on their heights or on the exponent spans has been asserted. □

5 A finite entropy estimate for weakly separated templates

For $z \in \mathbb{Z}_3$ let $[z]_j \in \{0, \dots, 3^j - 1\}$ denote its residue modulo 3^j, and define the length-q window

$$B_j^{(q)}(z) = \frac{[z]_{j+q} - [z]_j}{3^j}.$$

Let $\Sigma_L = \{\sum_{j=0}^{L-1} x_j 3^j : x_j \in \{0, 1\}\}$. All entropies and logarithms below use the natural logarithm.

We first specify the external finite result. This is Appendix A, "Exact theorem", of [4].

Theorem 5.1 (Finite causal-container input)**.** *Let $r \geq 3$, $V_0 = \{0, \dots, N - 1\}$, and Ω be an alphabet of size Q. A nonempty finite family $\mathcal{W} \subseteq \mathcal{X}^L$ determines letters $Z_j(x) \in \Omega$; these maps need not be local. Let a fixed family of ordered positional edges $e = (j_0 < \cdots < j_{r-1})$ have relations $\mathcal{R}_e(x_{<j_0}) \subseteq \Omega^r$ depending only on the indicated input prefix, and suppose every word satisfies every relation. Assume that distinct edges with the same source have disjoint future position sets and that each unordered pair of positions belongs to at most C edges.*

For positive integer thresholds $T_{\mathrm{s}}, T_2, \dots, T_{r-1}$ put $D = CQ^{r-3}$ and

$$\theta = Q \left[\frac{D \sum_{j=2}^{r-1} T_j + 1}{T_{\mathrm{s}}} + \frac{1}{T_2} + D \sum_{j=3}^{r-1} \frac{1 + \sum_{l=2}^{j-1} T_l}{T_j} \right]. \tag{16}$$

If $\theta \leq 1/2$, each word has a fingerprint $F(x)$ and nonempty assigned alphabets $A_j(x) \subseteq \Omega$ such that $Z_j(x) \in A_j(x)$, $A_j(x)$ is determined by $(x_{<j}, F(x))$, and, outside at most

$$Q \left(T_{\mathrm{s}} + D \sum_{j=2}^{r-1} T_j \right) N \tag{17}$$

edges, the full Cartesian product *of their assigned alphabets is contained in* $\mathcal{R}_e(x_{<j_0})$. *For every distribution on* $\mathcal{W}$,

$$H(F) \le N\big[h_{\mathrm{bin}}(\theta) + \theta \log Q\big], \quad h_{\mathrm{bin}}(t) = -t\log t - (1-t)\log(1-t).$$

The proof of Theorem 5.1 is reproduced in Appendix A from the cited frozen source. The remaining argument, including its application to unrestricted ternary inputs and rational units, is given here.

Proposition 5.2 (Finite-word bound)**.** *Choose* $b \ge 1$ *so that* $3^b > 2(r-1)$ *and put* $q = b+1$. *Suppose* (1) *has successive position gaps at least this* q *and pairwise disjoint positive distance sets, as in Theorem 2.1. Set*

$$q = b+1, \quad Q = 3^{(r-1)q}, \quad C = \binom{r}{2}, \quad D = CQ^{r-3}, \quad R = \max_i E_i.$$

For $L \ge q$ *let* W_L *be the set of all ternary input words modulo* 3^L *for which* $[c_i X]_L \in \Sigma_L$ *for every* $1 \le i \le K$*; no infinite extendability is required. Set* $N = L - q + 1 = L - b$. *With thresholds as in Theorem 5.1, define*

$$\epsilon_F = h_{\mathrm{bin}}(\theta) + \theta \log Q, \qquad \eta = \frac{Q(T_{\mathrm{s}} + D\sum_{j=2}^{r-1} T_j)}{K}.$$

If $\theta \le 1/2$, *then*

$$\log |W_L| \le N\left[\epsilon_F + \frac{\log 2}{r} + \eta \log 3\right] + (R+b)\log 3. \tag{18}$$

5.1 Ordinary carries and the positional graph

For a source j and a template i with $v = j + E_i < N$, consider the r window positions

$$j, \quad v - e_{r-2,i}, \ldots, v - e_{1i}, \quad v.$$

At these positions use respectively the windows of X, the corresponding $A_t X$, and X again. The joint letter available at every position is

$$Z_j(X) = (B_j^{(q)}(X), B_j^{(q)}(A_1 X), \ldots, B_j^{(q)}(A_{r-2}X)). \tag{19}$$

It takes values in an alphabet of size Q.

At position v, the identity for $c_i X$ has r summands: X, the $r-2$ terms $3^{e_{ti}} A_t X$, and $3^{E_i} u_i X$. Their ordinary nonnegative residues below position v sum to a carry $h \in \{0, \ldots, r-1\}$ into that position. The terminal summand has window, modulo 3^q,

$$u_i B_j^{(q)}(X) + D_{ij}(X_{<j}), \qquad D_{ij} = \frac{3^{E_i} u_i [X]_j - [3^{E_i} u_i [X]_j]_v}{3^v}.$$

Here $D_{ij} \in \mathbb{Z}_3$ depends only on the input digits before j. Consequently the corresponding edge relation is

$$u_i B_j^{(q)}(X) + \sum_{t=1}^{r-2} B_{v-e_{ti}}^{(q)}(A_t X) + B_v^{(q)}(X) + D_{ij} + h \in \Sigma_q \pmod{3^q} \tag{20}$$

for some $0 \le h \le r-1$. This derivation is valid for rational 3-adic units, with all residues interpreted in $\mathbb{Z}_3$. It imposes no restriction $X \in \Sigma$.

Every future displacement from j is a positive distance in P_i. Thus different templates have disjoint future sets at the same source. If two positions belong to an edge, their distance belongs to exactly one of the template distance sets, and there are at most $\binom{r}{2}$ choices of roles in that template. The source is then fixed. The graph therefore satisfies the two hypotheses of Theorem 5.1 with the stated C. The relations (20) are causal, as their only extra parameter is D_{ij}.

5.2 The one-digit Cartesian-product bound

Freeze the low $b = q-1$ digits of each of the r windows. Let each remaining high digit range in a nonempty subset of $\mathbb{F}_3$. Suppose the full Cartesian product of these choices satisfies (20), allowing the target and carry to vary with the choice. For a target $y \in \Sigma_q$, its low part satisfies $0 \le [y]_b \le R_b = (3^b - 1)/2$. Hence

$$[y]_b - h \in [-(r-1), R_b].$$

This interval has length $R_b + r - 1 < 3^b$. For the fixed residue modulo 3^b determined by the frozen inputs and D_{ij}, it therefore contains at most one representative. After subtracting that representative and dividing by 3^b, the sum of the high-digit factors, with the first multiplied by the unit u_i mod 3, lies in a single translate of $\{0,1\} \subset \mathbb{F}_3$.

No factor can have size three. Two factors of size at least two would already have sumset $\mathbb{F}_3$. Thus at most one factor has size two and all the others have size one. The product of the factor cardinalities is at most two, or equivalently their log-cardinalities sum to at most $\log 2$.

5.3 Conditional supports, incidence, and entropy

Apply Theorem 5.1 to the uniform distribution on W_L, using (19) and (20). (The set is nonempty because it contains zero.) For each realized word and $0 \le j < N$ put

$$C_j = \mathrm{supp}(X_{j+b} \mid X_{<j+b}, F), \qquad w_j = \log |C_j| \le \log 3.$$

Under this conditioning the low b digits of every window of X or $A_t X$ starting at j are fixed. Multiplication by a unit maps the high input digit bijectively onto the high output digit, so the projected support of the appropriate window has cardinality $|C_j|$. Moreover it lies in the relevant projection of the assigned alphabet A_j: every word in this finer conditional atom has the same $(X_{<j}, F)$ as the realized word.

On a good edge the full-product conclusion of Theorem 5.1 permits these projected supports to be combined, although the supports at different positions come from different prefix atoms. Freezing the low digits to their realized values and using the preceding Cartesian-product bound gives

$$\sum_{j \in e} w_j \le \log 2.$$

A bad edge instead costs at most $r \log 3$. Let E_{bad} be the number of bad edges for the realized word. There are at most KN edges. For a given template, each role covers all positions except at most $E_i \le R$ boundary positions. Summing incidences over all roles and templates gives

$$rK \sum_{j=0}^{N-1} w_j - rKR \log 3 \le \sum_e \sum_{j \in e} w_j \le KN \log 2 + rE_{\mathrm{bad}} \log 3.$$

The estimate remains valid when some templates have no edges because $E_i \ge N$, since the stated boundary loss then dominates the missing incidences. Dividing by rK and using $E_{\mathrm{bad}} \le \eta KN$ yields

$$\sum_j w_j \le N \frac{\log 2}{r} + \eta N \log 3 + R \log 3.$$

The windows' high input digits are precisely the positions $b, b+1, \ldots, L-1$. The first b digits cost at most $b \log 3$. The chain rule and the support bound on conditional entropy give

$$H(X \mid F) \le b \log 3 + \mathbb{E} \sum_j w_j.$$

Finally $H(X) \le H(F) + H(X \mid F)$, and the fingerprint bound is $H(F) \le N\epsilon_F$. Since X is uniform on W_L, this proves (18) and Proposition 5.2.

6 Compact covers and the limit of finite-visit dimensions

Fix $r \ge 3$. Choose b, $q = b+1$, Q, C, D as above. For an integer $u \ge 2$ set

$$T_j = u^{2^{j-1}-1} \ (2 \le j \le r-1), \qquad T_{\mathrm{s}} = u^{2^{r-1}-1}, \qquad K = u^{2^r}. \tag{21}$$

Every ratio occurring in (16) tends to zero as $u \to \infty$. Also $Q(T_{\mathrm{s}} + D\sum_j T_j)/K \to 0$. Thus

$$\theta \longrightarrow 0, \qquad \epsilon_F \longrightarrow 0, \qquad \eta \longrightarrow 0 \quad \text{for fixed } r. \tag{22}$$

For any one extracted finite template system $\mathcal{D}$ define

$$K_{\mathcal{D}} = \{X \in \mathbb{Z}_3 : c_i X \in \Sigma \text{ for } 1 \le i \le K\}.$$

It is compact. Its number of occupied length-L cylinders is at most $|W_L|$. Proposition 5.2, applied with the fixed span R of this system and then $L \to \infty$, gives

$$\overline{\dim}_{\mathrm{B}} K_{\mathcal{D}} \le s_{r,u} := \frac{\log 2}{r \log 3} + \frac{\epsilon_F}{\log 3} + \eta. \tag{23}$$

The fact that W_L contains all locally valid finite words only makes this upper bound larger; an assumption of infinite extendability is unnecessary.

Proof of Theorem 1.1. Given $\delta > 0$, first choose r so that $\log 2/(r \log 3) < \delta/2$. Then choose a finite u sufficiently large that $\theta \le 1/2$ and $\epsilon_F/\log 3 + \eta < \delta/2$. Fix the associated K, q and $J = J(r, K, q)$ from Theorem 2.1.

For any $x \in F_J$, choose J distinct true visit slopes $\gamma = 2^n$ and apply that finite theorem. The normalized variable $X = \alpha x$ lies in $K_{\mathcal{D}}$, since $c_i X = \gamma_i x \in \Sigma$. The normalized input X need not lie in Σ, which is why Proposition 5.2 was stated for arbitrary ternary inputs.

All coefficients constructed in the extraction are rational, and all exponents and visit times are ordinary integers. There are only countably many possible finite systems $\mathcal{D}$, including their unit anchors $\alpha_{\mathcal{D}}$. Hence

$$F_J \subseteq \bigcup_{\mathcal{D}} \alpha_{\mathcal{D}}^{-1} K_{\mathcal{D}}.$$

These are compact sets, and multiplication by a 3-adic unit is an isometry. By (23), each has upper box dimension at most $s_{r,u} < \delta$. This proves the covering statement.

Packing dimension is countably stable and is bounded above by upper box dimension on bounded sets. Therefore

$$\dim_{\mathrm{H}} F_J \le \dim_{\mathrm{P}} F_J \le \delta.$$

As $F_{k+1} \subseteq F_k$ and $\delta > 0$ was arbitrary, both limiting dimensions are zero. □

Remark 6.1 (Fixed finite-rank groups). The same proof applies to any fixed subgroup $G \le \mathbb{Q}^\times \cap \mathbb{Z}_3^\times$ of finite rank ρ. Replace distinct powers of two by distinct members $\gamma_i \in G$ and put $F_k(G) = \{x \in \mathbb{Z}_3 : \#\{\gamma \in G : \gamma x \in \Sigma\} \ge k\}$. In Lemma 2.2 the variable group has rank at most $\rho + d$, so replace the factor $d+2$ in B_d by $\rho + d + 1$. The normalized solution tuples are still distinct: equality of their first coordinates gives $\gamma_i/\gamma_j = 3^{e_{t_0 i} - e_{t_0 j}}$; the left side is a unit, so the exponent difference is zero and $\gamma_i = \gamma_j$. All remaining arguments are unchanged. The coefficients stay rational, and the compact-cover data are countable. Thus for each $\delta > 0$ some finite $J = J(\delta, \rho)$ gives the same compact-cover conclusion for $F_J(G)$, and both finite-visit dimension limits are zero. The finite-rank hypothesis is essential to this argument; this remark does not concern arbitrary families of integer multipliers.

Corollary 6.2. *The infinite-visit exceptional set*

$$E_2 = \{x \in \mathbb{Z}_3 : 2^n x \in \Sigma \text{ for infinitely many } n \geq 0\}$$

has Hausdorff and packing dimension zero.

Proof. $E_2 \subseteq F_J$ for every J, so the result follows from Theorem 1.1. □

The finite-visit conclusion supplies uniformity in the number of visits, rather than a statement about a particular initial point. It does not imply that $1 \notin E_2$, that every nonzero orbit has finitely many visits, or that the exceptional set is countable. In particular no resolution of the pointwise Erdős problem is claimed. The effective constants supplied by repeated Ramsey numbers and the unit-equation bound are far too large for the result to serve as a practical search cutoff.

7 The real exceptional set and the proof dependencies

Define

$$\mathcal{C}_{\mathbb{R}} = \left\{ \sum_{j=-\infty}^{m} \varepsilon_j 3^j : m \in \mathbb{Z},\ \varepsilon_j \in \{0,1\} \right\} \subseteq [0,\infty),$$

$$E_{\mathbb{R}} = \{\lambda > 0 : 2^n \lambda \in \mathcal{C}_{\mathbb{R}} \text{ for infinitely many } n \geq 0\}.$$

The digit condition includes every fractional digit. This is the untruncated real exceptional set in Lagarias [1, §1.2], whose Conjecture 1.4 asks whether $\dim_{\mathrm{H}} E_{\mathbb{R}} = 0$. The corresponding statement is called Conjecture A in the arXiv manuscript [5]. It differs from the truncated set defined using only $\lfloor 2^n \lambda \rfloor$: Lagarias [1, Theorem 1.3] proved that the latter has Hausdorff dimension $\log_3 2$.

Theorem 7.1. *The untruncated real exceptional set satisfies*

$$\boxed{\dim_{\mathrm{H}} E_{\mathbb{R}} = \dim_{\mathrm{P}} E_{\mathbb{R}} = 0.}$$

More precisely, for every $\delta > 0$ there is a countable cover of $E_{\mathbb{R}}$ by compact subsets of $\mathbb{R}$ of upper box dimension at most δ.

The real proof uses the same finite causal-container theorem, Theorem 5.1, whose complete frozen proof from version 1.0.0 is reproduced in Appendix A. Its two-translate capacity estimate is Lemma 9.1 below; we reproduce its proof from [4, Lemma 2.1] as well. Thus neither finite input is left implicit. We give the full affine finite-word application, the transfer to real prefixes, and the dimension argument. We also restate and prove the arithmetic extraction for an arbitrary infinite set of visit times. Its external input is the finite-rank unit-equation theorem of Evertse, Schlickewei and Schmidt [3]. This infinite extraction uses a true visiting anchor; it is distinct from the uniform finite extraction with a virtual anchor in Theorem 2.1.

Throughout, logarithms and Shannon entropies are natural. Write

$$\Sigma_L = \left\{ \sum_{j=0}^{L-1} \varepsilon_j 3^j : \varepsilon_j \in \{0,1\} \right\} \subseteq \mathbb{Z}/3^L\mathbb{Z}.$$

For $z \in \mathbb{Z}_3$, let $[z]_m$ be its representative modulo 3^m in $[0, 3^m)$, and put

$$B_{j,q}(z) = \left[(z - [z]_j)/3^j \right]_q .$$

A rational number with denominator prime to 3 is interpreted in $\mathbb{Z}_3$ when residues or valuations are used. Its ordinary real value is used in floors and real intervals. This distinction is essential in the transfer below.

8 Templates from an infinite set of exponents

The following extraction is the version of [4, Lemma 3.1] that separates the arithmetic from the space in which the initial value lies.

Lemma 8.1. *Let $S \subseteq \mathbb{Z}_{\geq 0}$ be infinite, and fix integers $r \geq 3$, $K \geq 1$ and $q \geq 1$. There exist distinct indices $n_0, n_1, \ldots, n_K \in S$ and ordinary integer units $A_1, \ldots, A_{r-2} \in \mathbb{Z}_3^\times$ such that $c_i = 2^{n_i - n_0}$ has an exact expansion*

$$c_i = 1 + \sum_{t=1}^{r-2} A_t 3^{e_{ti}} + u_i 3^{E_i}, \qquad u_i \in \mathbb{Z}_3^\times, \tag{24}$$

where $0 < e_{1i} < \cdots < e_{r-2,i} < E_i$, every successive gap, including the gap from 0, is at least q, and

$$\min\{e_{1,i+1}, e_{2,i+1} - e_{1,i+1}, \ldots, E_{i+1} - e_{r-2,i+1}\} > E_i \qquad (1 \leq i < K). \tag{25}$$

Proof. The distinct slopes $\alpha = 2^n$, $n \in S$, lie in the compact group $\mathbb{Z}_3^\times$. Choose a subsequence $\alpha_j \to \rho \in \mathbb{Z}_3^\times$. After discarding a possible equality, write

$$\rho^{-1}\alpha_j = 1 + v_{1j} 3^{e_{1j}}, \qquad v_{1j} \in \mathbb{Z}_3^\times, \quad e_{1j} \to \infty.$$

Pass to a subsequence on which $v_{1j} \to a_1 \in \mathbb{Z}_3^\times$. Whenever $v_{1j} - a_1 \neq 0$, normalize this difference by its valuation and take a further convergent subsequence of its unit part. Repeating this procedure produces successive exponent gaps tending to infinity.

An infinite collection of exact stops after s terms would give infinitely many solutions of

$$\rho^{-1}\gamma - \sum_{t=1}^{s} a_t 3^{e_t} = 1, \qquad \gamma \in \langle 2 \rangle, \quad 0 < e_1 < \cdots < e_s.$$

The coordinate variables belong to the finite-rank multiplicative group $\langle 2 \rangle \times \langle 3 \rangle^s$. Every nonempty subsum on the left is nonzero: its terms have distinct 3-adic valuations $0, e_1, \ldots, e_s$, so the term of least valuation cannot cancel. The characteristic-zero finite-rank theorem of [3], applied over an algebraic closure of $\mathbb{Q}_3$, excludes infinitely many such nondegenerate solutions. Exact stops may therefore be discarded at each of the finitely many stages required here.

We obtain

$$\rho^{-1}\alpha_j = 1 + \sum_{t=1}^{r-2} a_t 3^{e_{tj}} + u_j^* 3^{E_j}, \qquad a_t, u_j^* \in \mathbb{Z}_3^\times,$$

with every successive gap tending to infinity. Choose K members, now indexed by i, so that the required lower gap bounds and (25) hold. Choose ordinary integer units A_t with

$$v_3(A_t - a_t) \geq \max_i (E_i - e_{ti} + q).$$

Finally choose another member $\alpha_0 = 2^{n_0}$ of the original subsequence, outside the selected finite set, with

$$v_3(\rho/\alpha_0 - 1) \geq \max_i (E_i + q).$$

In replacing α_i/ρ by α_i/α_0, and a_t by A_t, the total error in row i is divisible by 3^{E_i+q} in $\mathbb{Z}_3$. The normalized final remainder is consequently still a unit. This gives (24) with the same exponents and separation. □

Remark 8.2. The slopes c_i are positive rational 3-adic units, but need not be ordinary integers: the selected anchor n_0 can exceed some n_i. The proof used only the infinite set S, not an initial point in $\mathbb{Z}_3$. In particular, it applies to true visit times of a real point.

9 An affine finite-word estimate

Fix a template system as in Lemma 8.1. Choose $k \geq 1$ and $b \geq 1$ with $3^b > r-1$, and use $q = b + k$ in the extraction. Set

$$R = \max_i E_i, \qquad Q = 3^{(r-1)q}, \qquad C = \binom{r}{2}, \qquad D = CQ^{r-3}.$$

For positive integer thresholds $T_{\mathrm{s}}, T_2, \ldots, T_{r-1}$ define

$$\theta = Q\left[\frac{D\sum_{j=2}^{r-1} T_j + 1}{T_{\mathrm{s}}} + \frac{1}{T_2} + D\sum_{j=3}^{r-1} \frac{1 + \sum_{l=2}^{j-1} T_l}{T_j}\right], \tag{26}$$

$$\epsilon_F = h_{\mathrm{bin}}(\theta) + \theta \log Q, \qquad h_{\mathrm{bin}}(t) = -t\log t - (1-t)\log(1-t), \tag{27}$$

$$\eta = \frac{Q(T_{\mathrm{s}} + D\sum_{j=2}^{r-1} T_j)}{K}, \qquad s = \frac{\epsilon_F}{\log 3} + \frac{1}{k} + \frac{2^{k+1}\log 2}{rk\log 3} + \eta. \tag{28}$$

We assume $\theta \leq 1/2$.

The following estimate is reproduced from [4, Lemma 2.1]. It is uniform in both target translates.

Lemma 9.1 (Two-translate capacity)**.** *Suppose $A_1, \ldots, A_r$ are nonempty subsets of $\mathbb{Z}/3^k\mathbb{Z}$ and*

$$T = A_1 + \cdots + A_r \subseteq (z_1 + \Sigma_k) \cup (z_2 + \Sigma_k).$$

Then

$$\sum_{i=1}^{r} \log|A_i| \leq r\log 3 + 2^{k+1}\log 2 =: B_{r,k}. \tag{29}$$

Proof. Let $H = \mathrm{Stab}(T)$ and $|H| = 3^h$. A coset of H fixes the lowest $k-h$ digits. Its intersection with a translate of Σ_k has at most 2^h elements: after translating back, the highest h digits form a word of Σ_h. Since the nonempty H-periodic set T contains a whole coset, $3^h \leq 2^{h+1}$. Thus $h \leq 1$.

Set $n_i = |A_i + H|/|H|$. Kneser's addition theorem [6], applied in the quotient by H, gives

$$\sum_i (n_i - 1) \leq |T|/|H| - 1 \leq 2^{k+1}/|H| - 1.$$

Indeed the final sum in the quotient has trivial stabilizer; the stabilizer of any partial sum is contained in that of the final sum, so the two-summand aperiodic inequality may be applied successively. Since $n \leq 2^{n-1}$ for positive integers n,

$$\sum_i \log|A_i| \leq rh\log 3 + \sum_i (n_i - 1)\log 2 \leq r\log 3 + 2^{k+1}\log 2.$$

□

Proposition 9.2. *For arbitrary fixed shifts $\beta_1, \ldots, \beta_K \in \mathbb{Z}_3$ and $L \geq q$, let*

$$W_L(\boldsymbol{\beta}) = \{x \in \{0, \ldots, 3^L - 1\} : [x]_L \in \Sigma_L,\ [c_i x + \beta_i]_L \in \Sigma_L\ (1 \leq i \leq K)\}.$$

Then, with $N = L - q + 1$,

$$\log|W_L(\boldsymbol{\beta})| \leq Ns\log 3 + (R + b + k - 1)\log 3 \tag{30}$$

whenever the set is nonempty. Equivalently, including the empty case,

$$|W_L(\boldsymbol{\beta})| \le 3^{R+b+k-1}\, 3^{sL}. \tag{31}$$

Both bounds are uniform in the shifts. All locally admissible finite words are counted; an extension to an infinite admissible word is not required.

Proof. We give the affine application in detail because uniformity in the shifts is needed for the real transfer. View a finite input word as its ordinary representative $X \in [0, 3^L)$, extended by zero digits if needed, and set $Y_t = A_t X$ in $\mathbb{Z}_3$.

Exact causal arithmetic. For template i and source position j, put $v = j + E_i$. The participating windows are

$$B_0 = B_{j,q}(X), \quad B_t = B_{v-e_{ti},q}(Y_t)\ (1 \le t \le r-2), \quad B_{\mathrm{end}} = B_{v,q}(X).$$

Write $P = [X]_j$ and define

$$T = \beta_i + 3^{E_i} u_i P, \qquad t_v = [T]_v, \qquad D_j = (T - t_v)/3^v \in \mathbb{Z}_3.$$

The center D_j depends only on the fixed data and the input prefix $X_{<j}$. In the exact identity for $c_i X + \beta_i$, the lower parts are

$$[X]_v, \qquad 3^{e_{ti}}[Y_t]_{v-e_{ti}}\ (1 \le t \le r-2), \qquad t_v.$$

These are r nonnegative integers below 3^v, so their ordinary carry lies in $\{0, \dots, r-1\}$. Taking the next q digits gives the relation

$$u_i B_0 + \sum_{t=1}^{r-2} B_t + B_{\mathrm{end}} + D_j + e \in \Sigma_q \pmod{3^q} \qquad \text{for some } 0 \le e \le r-1. \tag{32}$$

Only windows contained in the first L output digits will be used. Thus this relation is valid for every word in $W_L(\boldsymbol{\beta})$.

Low digits and capacity. On any Cartesian product of windows satisfying (32), fix the lowest b digits of each argument. The low residue of their sum, including D_j, is fixed. If $y \in \Sigma_q$ is an allowed output, then

$$y_{\mathrm{low}} - e \in [-(r-1), 3^b - 1].$$

Since $3^b > r-1$, a fixed residue modulo 3^b has at most two representatives in this interval. Dividing the corresponding identities by 3^b shows that the sum of the high k-digit factors lies in at most two translates of Σ_k. The first high factor is multiplied by the unit u_i, an affine bijection after its low digits are fixed. We use Lemma 9.1: for nonempty $H_1, \dots, H_r \subseteq \mathbb{Z}/3^k\mathbb{Z}$,

$$H_1 + \cdots + H_r \subseteq (z_1 + \Sigma_k) \cup (z_2 + \Sigma_k) \quad \Longrightarrow \quad \sum_{\nu=1}^{r} \log|H_\nu| \le r\log 3 + 2^{k+1}\log 2 =: B_{r,k}. \tag{33}$$

The proof above derives this estimate from Kneser's theorem; its constants do not depend on either translate.

The finite causal-container input. At each position $j \in \{0, \dots, N-1\}$ take the joint letter

$$Z_j = (B_{j,q}(X), B_{j,q}(Y_1), \dots, B_{j,q}(Y_{r-2}))$$

in an alphabet of size Q. The ordered positional edges are

$$(j, j + E_i - e_{r-2,i}, \dots, j + E_i - e_{1i}, j + E_i)$$

whenever all positions belong to $\{0,\dots,N-1\}$. Their relations select the indicated coordinates in (32). By (25), future position sets of distinct edges from the same source are disjoint. Every pair of positions belongs to at most $C=\binom{r}{2}$ edges: the distance belongs to at most one template, and that template has at most $\binom{r}{2}$ role pairs.

Theorem 5.1, with its complete proof in Appendix A, therefore supplies a deterministic fingerprint F and alphabets $\mathcal{A}_j$, each determined by $(X_{<j},F)$ and containing the actual joint letter. It gives

$$H(F)\le N\epsilon_F,\qquad E_{\mathrm{bad}}\le Q\left(T_{\mathrm{s}}+D\sum_{j=2}^{r-1}T_j\right)N. \tag{34}$$

Except for at most E_{bad} edges, the *full Cartesian product* of the assigned alphabets satisfies its relation at the actual source prefix. This full-product conclusion, rather than compatibility along one input, is the property used next.

Conditional supports and entropy. Take X uniformly distributed on the nonempty set $W_L(\boldsymbol{\beta})$. For each realized word put

$$C_j=\operatorname{supp}(X_{[j+b,j+b+k)}\mid X_{<j+b},F),\qquad w_j=\log|C_j|\le k\log 3.$$

Fixing $X_{<j+b}$ fixes the low b digits of all joint windows at j. Multiplication by any A_t induces an affine bijection on the next k digits. Each required high-coordinate projection of the conditional joint support consequently has exactly $|C_j|$ elements. This joint support is contained in $\mathcal{A}_j$, since $\mathcal{A}_j$ is determined by the shorter prefix $X_{<j}$ and the same fingerprint.

On a good edge these projections may be selected independently, even though the prefixes defining them differ at the different positions. The full-product property applies to every such selection; no common new input realizing all selections is asserted. The low-digit reduction and (33) imply, pointwise on that edge,

$$\sum_{\nu=0}^{r-1}w_{j_\nu}\le B_{r,k}.$$

On a bad edge use $rk\log 3$. Template i has $\max(N-E_i,0)$ edges, and each of its r roles misses $\min(E_i,N)$ positions. Accounting for these boundary incidences gives

$$rK\sum_{j=0}^{N-1}w_j\le B_{r,k}KN+rk\log 3\,E_{\mathrm{bad}}+rk\log 3\,KR,$$

and hence

$$\sum_j w_j\le\frac{NB_{r,k}}{r}+k\log 3\left(\frac{E_{\mathrm{bad}}}{K}+R\right). \tag{35}$$

For $h_t=H(X_t\mid X_{<t},F)$ the entropy chain rule yields

$$\sum_{t=j+b}^{j+b+k-1}h_t\le\mathbb{E}w_j.$$

The multiplicities of the h_t in this sum are at most k and total kN. Since $h_t\le\log 3$, their deficit from kL is at most $k(q-1)\log 3$ in entropy units. Therefore

$$H(X\mid F)\le k^{-1}\sum_j\mathbb{E}w_j+(q-1)\log 3.$$

Combining this with (34)–(35), and using $H(X)=H(F)+H(X\mid F)=\log|W_L(\boldsymbol{\beta})|$, proves (30). Dropping its nonpositive contribution $-(q-1)s\log 3$ gives (31). All bounds are independent of the shifts, as required. □

10 Transfer from real truncations to finite words

The next lemma is the additional step connecting the two number systems. It requires a bound for all finite words, not merely for prefixes that extend to infinite admissible 3-adic inputs.

Lemma 10.1 (Finite-word transfer)**.** *Let $c_i = p_i/q_i > 0$ be rational numbers in lowest terms, with $3 \nmid p_i q_i$, for $1 \le i \le K$. Suppose that, for some constants $C_0 > 0$, $s \ge 0$ and L_0, their affine word sets satisfy*

$$|W_L(\boldsymbol{\beta})| \le C_0\, 3^{sL} \quad (L \ge L_0)$$

for all $\boldsymbol{\beta} \in \mathbb{Z}_3^K$. For each integer $M \ge 1$, put

$$K_{\mathbb{R}}(\boldsymbol{c}; M) = \{X \in [0, M] : X \in \mathcal{C}_{\mathbb{R}},\ c_i X \in \mathcal{C}_{\mathbb{R}}\ (1 \le i \le K)\}.$$

This set is compact and has upper box dimension at most s.

Proof. On a bounded interval, there are only finitely many possible integer digit prefixes in $\mathcal{C}_{\mathbb{R}}$. Each has a compact set of fractional tails, the continuous image of the compact product $\{0,1\}^{\mathbb{N}}$. It follows that $\mathcal{C}_{\mathbb{R}}$ is closed on bounded intervals, and the stated finite intersection is compact.

Fix $X \in K_{\mathbb{R}}(\boldsymbol{c}; M)$ and $L \ge 1$. Define

$$T_L = \lfloor 3^L X \rfloor, \qquad T_{i,L} = \lfloor 3^L c_i X \rfloor, \qquad U_L = T_L \bmod 3^L \in \{0, \dots, 3^L - 1\}.$$

The full real digit conditions imply $U_L \in \Sigma_L$ and $T_{i,L} \bmod 3^L \in \Sigma_L$. Indeed, after multiplication by 3^L, the remaining fractional 0/1 tail lies in $[0, 1/2]$, so taking the floor gives exactly the allowed integer prefix. This also handles terminating expansions. Set

$$\beta_{i,L} = T_{i,L} - c_i T_L.$$

If $\tau = \{3^L X\}$ and $\tau_i = \{3^L c_i X\}$, then exactly

$$\beta_{i,L} = c_i \tau - \tau_i \in q_i^{-1}\mathbb{Z} \cap (-1, p_i/q_i) =: \mathcal{B}_i. \tag{36}$$

This is a fixed finite set of cardinality $p_i + q_i - 1$. Every element belongs to $\mathbb{Z}_3$ because $3 \nmid q_i$.

Now $T_L - U_L$ is an integer multiple of 3^L, so in $\mathbb{Z}_3$

$$c_i U_L + \beta_{i,L} = T_{i,L} - c_i(T_L - U_L) \equiv T_{i,L} \pmod{3^L}.$$

Consequently

$$U_L \in W_L(\beta_{1,L}, \dots, \beta_{K,L}). \tag{37}$$

The vector of shifts may change with X and with L. We take the finite union over $\prod_i \mathcal{B}_i$ at each depth; no consistency of a single shift vector across depths is needed.

For a fixed integer part of X, the value U_L determines a real half-open cell of length 3^{-L}. The first L fractional digits form the ordinary base-three integer U_L in their usual written order; the associated 3-adic word reads these same digits from low to high. Both orders preserve the condition that every digit is 0 or 1. There are at most $M + 1$ integer parts. Thus the number of occupied real cells of this length is at most

$$(M+1)\left(\prod_{i=1}^{K}(p_i + q_i - 1)\right) C_0\, 3^{sL}, \qquad L \ge L_0. \tag{38}$$

Closed intervals give a cover with the same bound up to a fixed factor. The factors preceding 3^{sL} are independent of L, so the upper box dimension is at most s. The discrete scales 3^{-L} suffice, since consecutive scales have bounded ratio. □

Remark 10.2. The variability of the shifts is real, even for fully admissible numbers. For $X = 1/2$ and $c = 1/4$, both X and $cX = 1/8$ belong to $\mathcal{C}_{\mathbb{R}}$, while (36) equals $-1/4$ for odd L and 0 for even L. Applying the container theorem after fixing each vector in the finite union is what preserves prefix dependence. The proof does not feed an unread real tail into a causal relation.

11 Countable compact covers and the dimension conclusion

Proof of Theorem 7.1. Fix $\delta > 0$. Choose first k sufficiently large, and then r so large that

$$\frac{1}{k} + \frac{2^{k+1} \log 2}{rk \log 3} < \delta.$$

Leave a positive margin in this inequality. Choose b with $3^b > r - 1$, and set $q = b + k$, Q, C, D as in Section 9. For integers $u \to \infty$ use the thresholds and number of templates

$$T_j = u^{2^{j-1}-1} \quad (2 \le j \le r-1), \qquad T_{\mathrm{s}} = u^{2^{r-1}-1}, \qquad K = u^{2^r}.$$

Every ratio in (26) tends to zero. Thus $\theta, \epsilon_F, \eta \to 0$, and we can fix one finite u with $\theta \le 1/2$ and $s \le \delta$ in (28). All these parameters are now fixed.

For $\lambda \in E_{\mathbb{R}}$ the true visit set

$$S(\lambda) = \{n \ge 0 : 2^n \lambda \in \mathcal{C}_{\mathbb{R}}\}$$

is infinite. Apply Lemma 8.1 to this set, obtaining true indices $n_0, n_1, \ldots, n_K$ and finite template data. Then $X = 2^{n_0}\lambda$ and $c_i X = 2^{n_i}\lambda$ all belong to $\mathcal{C}_{\mathbb{R}}$. For some positive integer M, therefore,

$$X \in K_{\mathbb{R}}(\boldsymbol{c}; M).$$

By Proposition 9.2 and Lemma 10.1, this compact set has upper box dimension at most $s \le \delta$.

Let $\mathcal{D}$ range over the finite data consisting of the visit indices, the integer exponents and the ordinary integer coefficients in Lemma 8.1. There are only countably many such data. We have proved the cover

$$E_{\mathbb{R}} \subseteq \bigcup_{\mathcal{D}} \bigcup_{M \ge 1} 2^{-n_0} K_{\mathbb{R}}(\boldsymbol{c}(\mathcal{D}); M). \tag{39}$$

No 3-adic accumulation point or limiting coefficient indexes this union. Each set is compact, and multiplication by 2^{-n_0} is a real similarity, so its upper box dimension is at most δ.

The span R and the rational heights p_i, q_i may be unbounded across the cover. This causes no problem: in (38) the depth L tends to infinity for one fixed finite system at a time. All of those quantities affect only its multiplicative counting constant. Countable stability of packing dimension and the usual inequalities $\dim_{\mathrm{H}} \le \dim_{\mathrm{P}} \le \overline{\dim}_{\mathrm{B}}$ on each compact component now imply

$$\dim_{\mathrm{H}} E_{\mathbb{R}} \le \dim_{\mathrm{P}} E_{\mathbb{R}} \le \delta.$$

Since $\delta > 0$ was arbitrary, both dimensions are zero. □

The real argument uses infinitely many visits in its extraction step; no uniform finite-visit conclusion for the real sets is asserted here. The 3-adic finite-visit conclusion of Theorem 1.1 was proved separately using the uniform finite extraction in Theorem 2.1. A zero-dimensional exceptional set need not be countable or empty. In particular, Theorem 7.1 does not settle whether $1 \in E_{\mathbb{R}}$, the remaining pointwise question associated with the powers of two.

The positive-dimensional truncated exceptional set does not contradict the transfer: a condition only on $\lfloor 2^n \lambda \rfloor$ gives no restriction on arbitrarily long fractional prefixes. In that setting the membership assertion (37) is unavailable.

12 Exact checks for the real transfer

The real proof consists of the arguments above, the reproduced finite causal-container proof in Appendix A, and the published unit-equation and sumset theorems [3, 6]. The finite inputs originating in version 1.0.0 are explicitly identified. The accompanying exact-arithmetic program checks the new finite-word bridge using rational numbers, including terminating and periodic ternary expansions. It verifies the bounds on the shifts, reduction with rational unit denominators, and membership of the resulting finite words. The current finite test set contains 1845 rational inputs and 49 slopes, with 632835 checked identities, including 12215 instances with fully admissible input and output expansions. Negative and nonintegral shifts are included, as is the alternating-shift example in Remark 10.2.

These computations detect errors in the finite identities; they do not prove the dimension theorem or replace the causal-container argument. ChatGPT was used in developing and editing the manuscript, implementing the exact checks, and conducting internal AI-assisted reviews from several mathematical perspectives. Such reviews do not constitute external peer review or proof-assistant verification. Authorship of this manuscript is attributed solely to Timofei Iuzhakov.

13 Digital rigidity and generic absence of three visits

Write

$$\Sigma = \left\{ \sum_{j\ge 0} e_j 3^j : e_j \in \{0,1\} \right\} \subset \mathbb{Z}_3, \qquad G = 1 + 3\mathbb{Z}_3, \qquad A = \Sigma \cap \mathbb{Z}_3^\times = 1 + 3\Sigma.$$

For a unit $b \in \mathbb{Z}_3^\times$, define

$$E_b^{(3)} = \{\lambda \in \mathbb{Z}_3 : b^n \lambda \in \Sigma \text{ for at least three distinct } n \ge 0\},$$
$$E_b = \{\lambda \in \mathbb{Z}_3 : b^n \lambda \in \Sigma \text{ for infinitely many } n \ge 0\}.$$

All Haar measures on compact groups are normalized to have mass one.

Theorem 13.1. *For Haar-almost every $b \in \mathbb{Z}_3^\times$,*

$$E_b^{(3)} = \{0\}.$$

In particular, $E_b = \{0\}$ for Haar-almost every unit b. The exceptional null set is independent of both the point λ and the three visit times.

This is a varying-base statement. The fixed-base exceptional sets were studied by Lagarias [1]; in particular, his Theorem 1.6(iii) (Theorem 1.5(3) in the arXiv version 4) gives $\dim_H E_2^{(3)} \ge (\log_3 2)/6$. Thus the conclusion of Theorem 13.1 cannot be imposed on the prescribed base 2. We do not attribute a varying-base conjecture to Lagarias. Our proof uses a finite classification of extremal digital planes, followed by an exact averaging identity over complete collections of prefixes. Both ingredients are proved below.

13.1 Full digital planes

For $m \ge 0$, put

$$D_m = \left\{ \sum_{j<m} e_j 3^j : e_j \in \{0,1\} \right\}, \qquad B_m = \left\{ \sum_{j<m} \sigma_j 3^j : \sigma_j \in \{-1,1\} \right\}, \qquad R_m = \frac{3^m - 1}{2}.$$

For unit coefficients $\alpha_1, \alpha_2, \alpha_3$ modulo 3^m, let

$$P_m(C;\alpha) = \#\{(X_1, X_2, X_3) \in B_m^3 : \alpha_1 X_1 + \alpha_2 X_2 + \alpha_3 X_3 \equiv C \pmod{3^m}\}.$$

At most three of the eight next-digit triples satisfy a prescribed residue modulo 3. Consequently $P_m(C;\alpha) \le 3^m$. We call a plane *full* if equality holds. Equivalently, every admissible node above the last level has three children. Coordinate sign changes, permutations, and simultaneous multiplication of the coefficients and target by a unit preserve the count.

Theorem 13.2 (Finite digital rigidity)**.** *For $m \ge 1$, a plane is full if and only if, after the operations just listed, its coefficients and target have one of the following forms modulo 3^m:*

$$(1,1,1), \qquad C \in B_m; \tag{a}$$

$$(1,1,1+\tau 3^g), \qquad C \in B_g, \quad 1 \le g < m, \quad \tau \in \{-1,1\}. \tag{b}$$

In (b), B_g means its 2^g ordinary integer representatives reduced modulo 3^m, not all lifts of B_g from modulus 3^g.

We first establish the coefficient coincidence that underlies the classification.

Lemma 13.3. *Normalize the coefficients by signs so that all three are 1 modulo 3. If a depth-m plane is full, two of these normalized coefficients are equal modulo 3^m.*

Proof. Divide by the first coefficient. Use binary coordinates through the bijection $X = 2x - R_m$ from D_m to B_m; modulo 3^m this is $X \equiv 2x + 1/2$. For fixed coefficients let F_l be the full binary targets at depth l. Reduction sends F_l into F_{l-1}, and each of the three admitted first-digit choices below a full target has a child target in F_{l-1}.

If all coefficients equal 1 modulo 3^m, there is nothing to prove. Otherwise let $k < m$ be their first unequal depth, so that

$$\alpha_i = 1 + 3^k a_i, \qquad a_1 = 0, \qquad a_1, a_2, a_3 \text{ are not all equal modulo 3.}$$

Here $k \ge 1$. For three equal binary coefficients, a target has three first children precisely when its first digit is 1 or 2; all three children have the same residual target. Induction gives

$$F_k = R_k + D_k.$$

For the canonical representative $V_0 \in F_k$, every admitted triple of depth-k prefixes satisfies $x_1 + x_2 + x_3 = V_0$ as an ordinary equality: the sum is at most $3R_k < V_0 + 3^k$, and a negative wraparound is impossible since $0 \le V_0 < 3^k$.

Consider a lift $V_0 + 3^k v$. Its next required residue, for an old prefix (x_1, x_2, x_3), is $v - \sum_i a_i (x_i \bmod 3)$. This residue must be nonzero. If $V_0 \equiv 1 \pmod 3$, the three first-digit choices exclude $v \in \{a_1, a_2, a_3\}$; if $V_0 \equiv 2 \pmod 3$, they exclude $v \in \{a_1 + a_2 + a_3 - a_j : 1 \le j \le 3\}$. All three choices occur in the full prefix tree. Three distinct a_i residues therefore permit no full lift. A split into two equal residues and one other permits exactly one lift. Since a full depth-m target exists, the latter case applies, and F_{k+1} is injective under reduction modulo 3^k. Fix the equal pair i, j; then $\alpha_i \equiv \alpha_j \pmod{3^{k+1}}$.

This target injectivity persists at every depth $l \ge k+1$. Indeed, suppose it holds at $l-1$, and take $V, V' \in F_l$ with the same reduction modulo 3^k. Their reductions modulo 3^{l-1} agree. Choose the same admitted first-digit triple for both targets. The child targets differ by $(V - V')/3$, which is zero modulo 3^{l-2}, and hence modulo 3^k when $l \ge k+2$. Injectivity in F_{l-1} gives $V \equiv V' \pmod{3^l}$.

If $m = k+1$, the selected coefficients already coincide modulo 3^m. Otherwise, among the three first children of a full depth-m target there are two whose targets differ by $\pm(\alpha_i - \alpha_j)/3$. They agree modulo 3^k. Injectivity in F_{m-1} forces their equality modulo 3^{m-1}, and therefore $\alpha_i \equiv \alpha_j \pmod{3^m}$. □

Proof of Theorem 13.2. By Lemma 13.3, we may work with balanced coefficients $(1, 1, t)$, where $t \equiv 1 \pmod 3$. If $t \equiv 1 \pmod{3^m}$, every admitted child map is $C \mapsto (C-\sigma)/3$, where $\sigma \in \{-1, 1\}$ represents C mod 3. The plane is full exactly when $C \in B_m$.

Otherwise set $g = v_3(t-1) < m$, using any unit lift of t to $\mathbb{Z}_3$. The child maps of a full state are

$$C \longmapsto \frac{C - \sigma t}{3} \quad \text{(twice)}, \qquad C \longmapsto \frac{C - \sigma(2-t)}{3} \quad \text{(once)}. \tag{40}$$

Put

$$\delta = \frac{2(1-t)}{t}, \qquad d \in \{-1, 1\}, \quad d \equiv \delta/3^g \pmod 3, \qquad \epsilon = -d, \qquad D = 1 - \epsilon 3^g.$$

For any full state of depth l, always following the first map in (40) shows that $z = C/t$ has balanced digits $\sigma_0, \ldots, \sigma_{l-1}$. Change only the branch at time j to the second map and then follow first maps again. The signs through time j are unchanged. Telescoping the digit expansions gives

$$z - \delta\sigma_j 3^j \in B_l \pmod{3^l}.$$

When $j + g < l$, the first affected digit is at $j + g$. It cannot be zero modulo 3, so

$$\sigma_{j+g} = \epsilon\sigma_j. \tag{41}$$

It follows, at every finite depth $l \geq g$, that

$$z \equiv P/D \pmod{3^l} \quad \text{for some } P \in B_g. \tag{42}$$

At $l = g$ this only records the first g digits, since $D \equiv 1$ modulo 3^g; at greater depths it is precisely (41). No infinite extension has been assumed.

Apply (42) to the two different children of the depth-m root. Their normalized targets satisfy

$$z_0 \equiv P_0/D, \qquad z_1 \equiv P_1/D \pmod{3^{m-1}}, \qquad P_0, P_1 \in B_g,$$

and, writing $\sigma = \sigma_0$,

$$P_1 - P_0 \equiv -\sigma\delta D/3 \pmod{3^{m-1}}. \tag{43}$$

This difference has valuation $g - 1$. Thus the two balanced words have equal lower $g - 1$ digits and opposite highest digits. The highest digit of P_0 is the root digit σ_g, which equals $\epsilon\sigma$ by (41). Therefore, as an ordinary integer equality,

$$P_1 - P_0 = -2\epsilon\sigma 3^{g-1}.$$

Equation (43) yields $\delta D \equiv 2\epsilon 3^g \pmod{3^m}$. The identity

$$\delta D - 2\epsilon 3^g = 2(D-t)/t$$

then gives $t \equiv D \pmod{3^m}$, and (42) at the root gives $C \in B_g$ at the full modulus 3^m. This also proves the boundary case $m = g + 1$: the two child words have length g, and their differing digit is still at position $g - 1$.

Conversely, take exact representatives $t = 1 - \epsilon 3^g$ and $C \in B_g$. The maps (40) become

$$\frac{C - \sigma}{3} \pm \epsilon\sigma 3^{g-1}.$$

Both remain in B_g: they shift its lower $g - 1$ remaining digits and insert an allowed highest sign. Every reachable state is a unit and has three children. The plane is full at all depths, and reduction gives (b). The common-coefficient case was already proved. □

For later use define the two-input count

$$\begin{aligned} Q_m(U;a,b) &= \#\{(x,y)\in D_m^2 : U+ax+by \bmod 3^m \in D_m\},\\ H_m(a,b) &= \max_{U \bmod 3^m} Q_m(U;a,b), \end{aligned} \tag{44}$$

where a, b are units. The balanced-coordinate conversion gives

$$Q_m(U;a,b) = P_m\big((a+b-1)/2 - 2U; (a,b,-1)\big). \tag{45}$$

In particular, $H_m(a,b) \le 3^m$.

Corollary 13.4. *If $a,b \in \mathbb{Z}_3^\times$ and $H_m(a,b) = 3^m$ for every $m \ge 1$, then*

$$a \in \{1,-1\}, \quad \text{or} \quad b \in \{1,-1\}, \quad \text{or} \quad a \in \{b,-b\}.$$

Proof. At every depth Lemma 13.3 gives one of these six equalities modulo 3^m. One equality occurs at unbounded depths, and hence holds in $\mathbb{Z}_3$. □

Corollary 13.5 (Enumeration and a rational height bound)**.** *There are exactly $24m-20$ ordered unit pairs (a,b) modulo 3^m for which $H_m(a,b) = 3^m$, and exactly $28\cdot 2^m - 48$ full triples (a,b,U). Moreover, let $a = A/D$, $b = B/D$, where A, B, D are ordinary integers prime to 3 and $\max(|A|,|B|,|D|) \le H$. If the pair is not full at every depth, then*

$$3^m > 4H^2 \quad \Longrightarrow \quad H_m(a,b) \le 3^m - 1.$$

Proof. Normalize a, b separately by signs to be 1 modulo 3. Theorem 13.2 and (45) give the normalized pairs

$$(1,1), \qquad (t,1),\ (1,t),\ (t^{-1},t^{-1}), \qquad t = 1\pm 3^g, \quad 1 \le g < m.$$

They are distinct: each nonconstant triple has a unique coincident pair, and $v_3(t-1) = g$ separates the levels. There are $6m-5$ normalized pairs and four sign choices. The common pair has 2^m targets per sign choice; each nonconstant pair of level g has 2^g. Hence the target count is $4\cdot 2^m + 24\sum_{g=1}^{m-1} 2^g = 28\cdot 2^m - 48$.

Suppose now $3^m > 4H^2$ and the rational pair is full at depth m. Finite coefficient coincidence makes one of $A \pm B$, $A \pm D$, $B \pm D$ divisible by 3^m. Its absolute value is at most $2H$, so it is zero. After normalization the triple is $(1,1,T/S)$ with $|S|, |T| \le H$. If $T = S$, the game is full at all depths. Otherwise put $g = v_3(T-S)$; then $3^g \le 2H < 3^m$. Classification gives $T - S(1\pm 3^g) \equiv 0 \pmod{3^m}$, while

$$|T - S(1\pm 3^g)| \le 2H + 2H^2 \le 4H^2 < 3^m.$$

This is again an exact equality, placing the coefficients in an all-depth full family. The assumed nonfullness proves the implication. □

13.2 Complete prefix cubes and Haar measure

We use the elementary power operation ρ^N, for $\rho \in G$ and $N \in \mathbb{Z}_3$. It is defined by the convergent binomial series, or equivalently by $\exp(N\log\rho)$. The series for logarithm and exponential give inverse group isomorphisms between G and $3\mathbb{Z}_3$. In particular a unit power is an automorphism of G. The coefficients $\binom{N}{k}$ are integral: approximate N by nonnegative integers in $\mathbb{Z}_3$ and use continuity of each binomial polynomial. These series also show that $\rho \mapsto \rho^N$ is 1-Lipschitz. We will use these facts only on principal units.

For fixed $N \in \mathbb{Z}_3 \setminus \{0,1\}$ define

$$\mathcal{B}_N = \{\rho \in G : \text{some } \mu \in A \text{ satisfies } \rho\mu, \rho^N\mu \in A\}.$$

This set is compact, being the projection of a closed subset of $G \times A$.

Proposition 13.6. *For every fixed $N \in \mathbb{Z}_3 \setminus \{0,1\}$, the set $\mathcal{B}_N$ has Haar measure zero in G.*

Proof. If $N \equiv 2 \pmod 3$, then $\mathcal{B}_N \subseteq \{1\}$. Indeed, write $\rho = 1 + 3^g t \neq 1$ with $g \geq 1$ and t a unit. The three numbers $\mu, \rho\mu, \rho^N\mu$ have equal digits below g, whereas their digits at g differ by $0, t, 2t$ modulo 3 because $\mu \equiv 1 \pmod 3$. One of these three digits is 2.

Next suppose $N = 3^s r$ with $s \geq 1$ and r a unit. Set

$$a(\rho) = (1-N)\rho^N, \qquad b(\rho) = r\rho^{N-1}, \qquad \Phi(\mu,\nu) = \mu^{1-N}\nu^N. \tag{46}$$

Both coefficients are units. A level-q node is a pair $(\mu,\nu) \in A_q^2$, where $A_q = 1 + 3D_{q-1}$, for which $\Phi(\mu,\nu) \bmod 3^q \in A_q$. Representatives in A_q are ordinary nonnegative integers. Its ratio is $\rho = \nu/\mu \in G$.

Fix $m \geq 1$ and $q \geq m+s+1$. Group pairs by fixing μ and the digits of ν below position $q-s$. Write the resulting pairs as

$$(\mu, \nu_e), \qquad \nu_e = \nu_0 + 3^{q-s}e, \qquad e \in D_s.$$

Every group containing a node contains all its 2^s possible nodes. More precisely, with $\rho_0 = \nu_0/\mu$,

$$\Phi(\mu,\nu_e) \equiv \Phi(\mu,\nu_0) + 3^q b(\rho_0)e \pmod{3^{q+m+1}}. \tag{47}$$

To prove this, expand in $3^{q-s}e/\nu_0$. From $\binom{N}{k} = (N/k)\binom{N-1}{k-1}$ we obtain $v_3\binom{N}{k} \geq s - v_3(k)$. For $k \geq 2$ the valuation of the kth term is at least

$$k(q-s) + s - v_3(k) \geq 2q - s \geq q + m + 1;$$

the first inequality follows from $v_3(k) \leq k-2$ and $q - s \geq 1$. This proves (47). In addition, all ratios in the group agree modulo 3^{m+1}.

Let U be the block of m digits of $\Phi(\mu,\nu_0)$ beginning at position q. The descendants of a group have the form

$$(\mu + 3^q x, \nu_e + 3^q y), \qquad x, y \in D_m.$$

Integral binomial coefficients and $2q \geq q+m+1$ give their exact new-block condition

$$U + be + ax + 3^s by \bmod 3^m \in D_m, \tag{48}$$

where a, b are evaluated at ρ_0. They can be used throughout the group by the Lipschitz property and the common ratio residue. The lower q digits remain admissible.

As $e \in D_s$ and $y \in D_m$ vary, $z = e + 3^s y$ runs bijectively over D_{m+s}. Each residue in D_m has exactly 2^s such lifts modulo 3^m, also when $m < s$. Summing (48) therefore gives the exact descendant count

$$\#\{\text{descendants of the group}\} = 2^s Q_m(U; a, b). \tag{49}$$

Fix a ratio class $\mathcal{C} \subset G$ modulo 3^{m+1}. The coefficients are constant modulo 3^m on this class. Write $H_{\mathcal{C}} = H_m(a,b)$, and let $T_q(\mathcal{C})$ count its level-q nodes. All the preceding groups stay in $\mathcal{C}$, as do their descendants. Since each group has 2^s original nodes, (49) implies

$$T_{q+m}(\mathcal{C}) \leq H_{\mathcal{C}} T_q(\mathcal{C}), \qquad q \geq m+s+1. \tag{50}$$

If $H_{\mathcal{C}} < 3^m$, iterate with $q_j = q_0 + jm$ to obtain $T_{q_j}(\mathcal{C}) \leq T_{q_0}(\mathcal{C}) H_{\mathcal{C}}^j$.

For a ratio ρ modulo 3^q, let $n_q(\rho)$ count the $\mu \in A_q$ for which $\rho\mu$ and $\rho^N\mu$ lie in A_q modulo 3^q. The substitution $\nu \equiv \rho\mu \pmod{3^q}$ is a bijection, and G has 3^{q-1} residue classes modulo 3^q. Consequently

$$\int_{\mathcal{C}} n_q(\rho)\, d\rho = \frac{T_q(\mathcal{C})}{3^{q-1}}.$$

Every $\rho \in \mathcal{B}_N$ has $n_q(\rho) \geq 1$ at each depth. It follows that

$$\mathrm{Haar}(\mathcal{B}_N \cap \mathcal{C}) \leq \frac{T_{q_j}(\mathcal{C})}{3^{q_j-1}} \longrightarrow 0 \qquad (H_{\mathcal{C}} < 3^m).$$

Taking the countable union over finite windows and ratio classes reduces the proof to the residual set

$$\mathcal{F}_N = \{\rho \in G : H_m(a(\rho), b(\rho)) = 3^m \text{ for every } m \geq 1\}.$$

In fact $\mathcal{F}_N$ contains at most two points. Let $\sigma \in \{-1, 1\}$ represent r mod 3. Since $\rho^N \equiv 1 \pmod{3^{s+1}}$,

$$a(\rho) - 1 \equiv -N \pmod{3^{s+1}}, \qquad v_3(a(\rho) - 1) = s.$$

Thus $a \neq 1$, and $a \equiv 1 \pmod 3$ also excludes $a = -1$. Corollary 13.4 leaves only $b = \sigma$ or $a = \sigma b$. The first has the unique solution $\rho_1 = (\sigma/r)^{1/(N-1)} \in G$, because $N - 1$ is a unit. The second, using $a/b = ((1 - N)/r)\rho$, has the unique solution $\rho_2 = \sigma r/(1 - N) \in G$. Hence $\mathcal{F}_N \subseteq \{\rho_1, \rho_2\}$, proving nullity. No assertion that either candidate is full is needed.

Finally, if $N \equiv 1 \pmod 3$, exchange μ and $\nu = \rho\mu$. The ratio becomes ρ^{-1} and the exponent becomes $1 - N$, which is divisible by 3 and different from $0, 1$. Thus inversion maps $\mathcal{B}_N$ into $\mathcal{B}_{1-N}$. Inversion preserves Haar measure, and the preceding case applies. □

13.3 One null set for all points and times

Proof of Theorem 13.1. For every nonzero integer e, the map $P_e : G \to G$ given by $P_e(\beta) = \beta^e$ has null preimages of Haar-null sets. Indeed, in logarithmic coordinates it is $x \mapsto ex$ on $3\mathbb{Z}_3$, an injective homomorphism onto an open subgroup. Its pushforward measure is normalized Haar measure on that subgroup, and is therefore absolutely continuous with respect to Haar measure on G.

By Proposition 13.6, the set

$$\mathcal{N} = \bigcup_{e \in \mathbb{Z}\setminus\{0\}} \bigcup_{N \in (\mathbb{Q}\cap\mathbb{Z}_3)\setminus\{0,1\}} P_e^{-1}(\mathcal{B}_N)$$

is Haar-null. Fix $\beta \in G \setminus \mathcal{N}$. If $\lambda \neq 0$ had three distinct visit times n_0, n_1, n_2, put $v = v_3(\lambda)$. Interchange n_1, n_2 if necessary so that

$$v_3(e) \leq v_3(e'), \qquad e = n_1 - n_0, \qquad e' = n_2 - n_0.$$

Then $N = e'/e \in (\mathbb{Q} \cap \mathbb{Z}_3) \setminus \{0, 1\}$, and the quantities

$$\mu = 3^{-v}\beta^{n_0}\lambda, \qquad \rho = \beta^e$$

satisfy $\mu, \rho\mu, \rho^N\mu \in A$. The identity $\rho^N = \beta^{e'}$ follows from logarithms. Thus $\rho \in \mathcal{B}_N$, contradicting the choice of β. This proves the theorem on G simultaneously for all nonzero λ and all triples of times.

For $b \equiv -1 \pmod 3$, the visit times of a fixed nonzero point have a common parity: the leading unit digit of $b^n\lambda$ must be 1. Three visits therefore give three visits of $b^{r_0}\lambda$ under the base $\beta = b^2 \in G$, for some $r_0 \in \{0, 1\}$. Squaring maps the coset $-1 + 3\mathbb{Z}_3$ bijectively onto G and has null preimages of null sets; write $b = -u$ and use the unit power automorphism $u \mapsto u^2$. This proves the statement on all units. Finally, 0 always visits Σ and $E_b \subseteq E_b^{(3)}$. □

Remark 13.7. The averaging in (49) is over a complete group of prefixes. It does not give a contraction on a prescribed ratio fiber. Accordingly, neither Theorem 13.1 nor the effective finite-window bound in Corollary 13.5 asserts finiteness of visits for a specified base and a specified point. The proofs above are independent of finite enumeration; the accompanying exact checks verify small instances of the classification and the complete-cube identity.

14 Summable denominators and positive-density multipliers

Write

$$\mathcal{D} = \left\{ \sum_{j=0}^{L-1} \epsilon_j 3^j : L \geq 1,\ \epsilon_j \in \{0,1\} \right\}, \qquad \Sigma = \left\{ \sum_{j \geq 0} \epsilon_j 3^j : \epsilon_j \in \{0,1\} \right\} \subset \mathbb{Z}_3.$$

The first sum is finite and defines an ordinary nonnegative integer; the second converges in the 3-adic metric. For a positive integer M set

$$\begin{aligned} C_M &= \{x \in \Sigma : Mx \in \Sigma\}, \\ \mathcal{M}_C &= \{M \geq 1 : \exists a \in \mathcal{D} \setminus \{0\},\ Ma \in \mathcal{D}\}, \\ \mathcal{M}_H &= \{M \geq 1 : \dim_H C_M > 0\}. \end{aligned}$$

Here Hausdorff dimension is taken in the 3-adic metric, and $M_C(X) = \#(\mathcal{M}_C \cap [1, X])$, with analogous notation for M_H. These are the two multiplier-counting questions in Problems 1 and 2 of the arXiv version of [1] (journal Problems 1.10 and 1.11). The nonzero requirement in $\mathcal{M}_C$ is essential.

Theorem 14.1. *There are constants $c_1, c_2 > 0$ such that, for all sufficiently large X,*

$$M_H(X) \geq c_1 X, \qquad M_H(X) - M_C(X) \geq c_2 X.$$

In particular $M_H(X) \asymp X$. This theorem does not assert that $M_H(X)/X$ has a limit, nor that this limit is $1/2$.

The proof uses reduced denominators in the ordinary middle-thirds set

$$\mathcal{C} = \left\{ \sum_{j \geq 1} 2\epsilon_j 3^{-j} : \epsilon_j \in \{0,1\} \right\}, \qquad s = \frac{\log 2}{\log 3}.$$

Define its set of internal reduced denominators prime to 3 by

$$\mathscr{D} = \{d > 1 : 3 \nmid d,\ \exists a,\ 0 < a < d,\ (a,d) = 1,\ a/d \in \mathcal{C}\}.$$

Thus endpoints and unreduced representations are not counted.

Theorem 14.2. *Put $\kappa = 4/3 - 2s > 0$. As $Y \longrightarrow \infty$,*

$$\sum_{\substack{d \in \mathscr{D} \\ d > Y}} \frac{1}{d} \ll (\log Y)^{-9} + Y^{-\kappa}.$$

In particular $\sum_{d \in \mathscr{D}} 1/d$ converges.

14.1 The carry obstruction

We give the carry argument explicitly; finite carry presentations of these intersections also appear in [7]. Suppose $M > 1$ and $M \equiv 1 \pmod 3$. Reading input and output digits $a, b \in \{0,1\}$ from low to high gives the transition

$$3c' = c + Ma - b, \tag{51}$$

starting at $c = 0$. The allowed pairs (a,b), according as $c \equiv 0, 1, 2 \pmod 3$, are respectively

$$\{(0,0),(1,1)\}, \qquad \{(0,1)\}, \qquad \{(1,0)\}.$$

All reachable carries are nonnegative. After t steps, if A, B are the input and output integers read so far, then $3^t c = MA - B$; since $A \le (3^t - 1)/2$ and $B \ge 0$, every reachable carry satisfies $c < M/2$. The reachable graph is therefore finite and has no dead ends. Conversely, an infinite path gives $Mx = y$ in $\mathbb{Z}_3$, because its bounded carry error is divisible by 3^t at every stage.

Lemma 14.3. *If $M > 1$, $M \equiv 1 \pmod 3$, and $\dim_H C_M = 0$, then there is an integer c with $0 \le 2c < M$ such that*

$$\frac{2c+1}{M+1} \in \mathcal{C} \cap (0,1).$$

Proof. If a reachable strongly connected component contains a vertex with two outgoing edges inside that component, choose closed paths U, V based there and starting with these different edges. Their first input digits differ. The return words UV, VU have the same length and different first input digits. Arbitrary concatenations of them, after a fixed path from the initial state, yield a binary block Cantor set of positive Hausdorff dimension. Thus no such component exists when $\dim_H C_M = 0$.

Choose a terminal strongly connected component. All its outgoing edges remain inside it, so its vertices cannot be 0 modulo 3. Each vertex has exactly one outgoing edge, and the component is a directed cycle. On this cycle $a + b = 1$. Put $q = M + 1$ and $r_j = (2c_j + 1)/q$. Equation (51) becomes $3r_{j+1} = r_j + 2a_j$. If the cycle length is L, iteration gives

$$r_0 = \frac{2\sum_{j=0}^{L-1} a_j 3^j}{3^L - 1} \in \mathcal{C}.$$

The inequalities on the carries give $0 < r_0 < 1$. □

Only this necessary condition for zero dimension will be used. A terminal cycle can coexist with a different branching component.

Lemma 14.4. $\mathcal{M}_C \subseteq \mathcal{M}_H$.

Proof. Let $Ma = b$ with positive $a, b \in \mathcal{D}$, and choose L at least the ternary lengths of both. All points $x = a\sum_{j\ge 0} \epsilon_j 3^{Lj}$, $\epsilon_j \in \{0,1\}$, lie in C_M: the input blocks are $0, a$ and the output blocks are $0, b$, without overlap. The set of the sums has dimension $\log 2/(L \log 3) > 0$; multiplication by the nonzero integer a is a 3-adic similarity and preserves dimension. □

14.2 A small-order estimate

For $n \ge 2$ with $(n, 3) = 1$ let $\mathrm{ord}_n(3)$ be the multiplicative order of 3 modulo n, and let $\lambda(n)$ be Carmichael's function. The only non-elementary input in this section is the following unconditional estimate, stated as Lemma 10 of [8]:

$$\#\{n \le X : \lambda(n) \le n \exp(-(\log\log n)^3)\} \ll \frac{X}{(\log X)^{10}}. \tag{52}$$

Finitely many small n can be omitted in this display. No hypothesis such as the generalized Riemann hypothesis is assumed.

Lemma 14.5.

$$\#\{2 \le n \le X : (n,3) = 1,\ \mathrm{ord}_n(3) < n^{1/3}\} \ll X(\log X)^{-10}.$$

Proof. Let $\mathscr{P}$ be the primes $p \neq 3$ for which $\mathrm{ord}_p(3) \leq p^{2/5}$. Every such $p \leq z$ divides $\prod_{1 \leq j \leq z^{2/5}} (3^j - 1)$. Taking logarithms gives $\#\{p \in \mathscr{P} : p \leq z\} = O(z^{4/5})$, and partial summation gives $\sum_{p \in \mathscr{P}} p^{-9/10} < \infty$.

Let $n_{\mathscr{P}}$ be the largest divisor of n supported on $\mathscr{P}$. For $f(n) = n_{\mathscr{P}}^{1/10}$, the multiplicative function $f * \mu$ is nonnegative. Hence

$$\sum_{n \leq X} f(n) \leq X \sum_{d \geq 1} \frac{(f * \mu)(d)}{d} = X \prod_{p \in \mathscr{P}} \frac{1 - p^{-1}}{1 - p^{-9/10}} \ll X.$$

It follows that $n_{\mathscr{P}} > X^{1/20}$ for at most $O(X^{199/200})$ integers $n \leq X$.

Similarly put $R(n) = n/\operatorname{rad}(n)$ and $h(n) = R(n)^{1/2}$. Here $(h * \mu)(p) = 0$, $(h * \mu)(p^e) \geq 0$, and $\sum_{e \geq 2} (h * \mu)(p^e)/p^e = O(p^{-3/2})$. The same argument gives $\sum_{n \leq X} h(n) \ll X$, so $R(n) > X^{1/20}$ for at most $O(X^{39/40})$ integers $n \leq X$.

We also need the elementary inequality

$$\mathrm{ord}_n(3) \geq \frac{\lambda(n)}{n} \prod_{p \mid n} \mathrm{ord}_p(3) \qquad ((n,3) = 1). \tag{53}$$

Indeed, for each $p^e \| n$ write $\lambda(p^e) = a_p b_p$ with $a_p = \mathrm{ord}_p(3)$. The integer $\mathrm{lcm}_p(a_p b_p)$ divides $(\prod_p b_p) \operatorname{lcm}_p a_p$, while $\operatorname{lcm}_p a_p$ divides $\mathrm{ord}_n(3)$ and $\prod_p \lambda(p^e) \leq n$. Rearrangement proves (53).

Consider $X/2 < n \leq X$, excluding the two moment exceptions above and the exceptions in (52). The product of the distinct prime factors of n outside $\mathscr{P}$ is at least

$$\frac{n}{n_{\mathscr{P}} R(n)} \geq \frac{1}{2} X^{9/10}.$$

Consequently (53) gives

$$\mathrm{ord}_n(3) \geq \exp(-(\log\log X)^3) \left(\tfrac{1}{2} X^{9/10}\right)^{2/5} > X^{1/3}$$

for sufficiently large X. In the last step one may use $\exp(-(\log\log X)^3) \geq X^{-1/100}$ eventually, since $9/25 - 1/100 = 7/20 > 1/3$. This proves the required estimate on each dyadic interval. Sum over the intervals between $\sqrt{X}$ and X, where their logarithms are comparable to $\log X$, and absorb the at most $\sqrt{X}$ remaining integers. □

14.3 Summability of Cantor denominators

Proof of Theorem 14.2. Distinct reduced fractions of denominator at most T are separated by at least T^{-2}. Cover $\mathcal{C}$ by its 2^k level-k intervals, where $k = \lfloor 2 \log_3 T \rfloor + 1$. Each interval has length $3^{-k} < T^{-2}$ and contains at most one of these fractions. Thus the number of reduced rational points of $\mathcal{C}$ with denominator at most T is $O(T^{2s})$.

For each $d \in \mathscr{D}$, the orbit of one reduced $a/d \in \mathcal{C}$ under $x \mapsto 3x \pmod 1$ stays in $\mathcal{C}$ and contains exactly $\mathrm{ord}_d(3)$ points. Orbits belonging to different reduced denominators are disjoint. It follows that

$$\#\{d \in \mathscr{D} : T/2 < d \leq T,\ \mathrm{ord}_d(3) \geq d^{1/3}\} \ll T^{2s - 1/3}.$$

Lemma 14.5 handles the remaining denominators, so

$$\#(\mathscr{D} \cap (T/2, T]) \ll T(\log T)^{-10} + T^{2s-1/3}.$$

Since $s < 2/3$ (equivalently $8 < 9$), we have $\kappa = 1 - (2s - 1/3) > 0$. Sum the reciprocal-denominator bound on the shells $(2^j Y, 2^{j+1} Y]$, $j \geq 0$. The power part is $O(Y^{-\kappa})$ and the logarithmic part is $O(\sum_{j \geq 0} (\log Y + j)^{-10}) = O((\log Y)^{-9})$. □

14.4 A positive-density sifted set

Let $\mathscr{B}$ be the odd members of $\mathscr{D}$ and put

$$\mathcal{Q} = \{q \geq 1 : q \equiv 5 \pmod 6,\ d \nmid q \text{ for every } d \in \mathscr{B}\}.$$

Lemma 14.6. *The set $\mathcal{Q}$ has a natural density $\delta > 0$, with*

$$\delta \geq \frac{1}{6} \prod_{d \in \mathscr{B}} (1 - 1/d) > 0.$$

Proof. For a finite subset $\mathscr{B}_0 \subset \mathscr{B}$, its avoidance set is periodic. All its moduli are coprime to 6. By the Chinese remainder theorem, conditioning on $q \equiv 5 \pmod 6$ leaves uniform, independent prime-power residue coordinates modulo $\operatorname{lcm}(\mathscr{B}_0)$. The truncated prime valuations are therefore independent. Each event $d \nmid q$ is decreasing in these valuations. Such decreasing events are positively associated. For completeness, in one coordinate, increasing or decreasing functions f, g in the same direction satisfy

$$2\operatorname{Cov}(f(U), g(U)) = \mathbb{E}[(f(U) - f(U'))(g(U) - g(U'))] \geq 0,$$

where U' is an independent copy. Conditioning on one coordinate and induction prove the same inequality in a product distribution. Applying it successively to products of decreasing indicators gives a density at least $\frac{1}{6}\prod_{d \in \mathscr{B}_0}(1 - 1/d)$. This is a lower bound; divisibility by different composite moduli is not being treated as independent.

Take $\mathscr{B}_0 = \mathscr{B} \cap [1, Y]$, and denote the density of this finite avoidance set by δ_Y. The densities decrease to some $\delta > 0$, since Theorem 14.2 makes the displayed infinite product positive. Uniformly in X, the count removed on imposing all the conditions with $d > Y$ is at most

$$\sum_{\substack{d \in \mathscr{B} \\ d > Y}} \lfloor X/d \rfloor \leq X \sum_{\substack{d \in \mathscr{B} \\ d > Y}} 1/d.$$

First let $X \to \infty$ with Y fixed, then let $Y \to \infty$. The tail tends to zero, proving that $\mathcal{Q}$ itself has density δ and establishing the claimed lower bound. □

14.5 Positive dimension and separation

Proof of Theorem 14.1. If $q \in \mathcal{Q}$, no internal fraction a/q belongs to $\mathcal{C}$: its reduced denominator would be an odd member $d > 1$ of $\mathscr{D}$ dividing q. Set $M = q - 1$. Then $M > 1$ and $M \equiv 1 \pmod 3$, so Lemma 14.3, by contraposition, gives $M \in \mathcal{M}_H$. Hence $M_H(X) \geq \delta X + o(X)$.

If a positive Cantor integer has highest nonzero ternary position j, its value belongs to $[3^j, (3^{j+1} - 1)/2]$. Consequently

$$\{b/a : a, b \in \mathcal{D} \setminus \{0\}\} \subseteq \bigcup_{j \in \mathbb{Z}} \left(\tfrac{2}{3}\, 3^j, \tfrac{3}{2}\, 3^j\right).$$

In particular no integer in $[3T/2, 2T]$ belongs to $\mathcal{M}_C$ when T is a power of 3. Choose such a T with $2T \leq X < 6T$. By Lemma 14.6, the interval $[3T/2 + 1, 2T + 1]$ contains $\delta T/2 + o(T)$ elements of $\mathcal{Q}$. Their translates $M = q - 1$ are in $(\mathcal{M}_H \setminus \mathcal{M}_C) \cap [1, X]$. For all sufficiently large X there are at least $\delta T/4 > \delta X/24$ of them. Use Lemma 14.4 to identify this set difference with the difference of the two counting functions. The trivial upper bound $M_H(X) \leq \lfloor X \rfloor$ completes the assertion $M_H(X) \asymp X$. □

Remark 14.7. The positive density proved above belongs to the explicit sifted subfamily $\{q - 1 : q \in \mathcal{Q}\}$, not necessarily to all of $\mathcal{M}_H$. The separation estimate is not a linear lower bound for $M_C(X)$. These results concern varying integer multipliers; they make no finiteness assertion about the visits of a fixed nonzero 3-adic point under multiplication by powers of 2.

15 The exact generalized exceptional dimension

The generalized exceptional set allows arbitrary positive integer multipliers in place of powers of two. Following [7, Definition 1.5], put

$$E_* = \{x \in \Sigma : Mx \in \Sigma \text{ for infinitely many } M \in \mathbb{Z}_{>0},\ 3 \nmid M\}.$$

For a nonzero $x \in \Sigma$, every such multiplier is congruent to 1 modulo 3, as is seen from the first nonzero digit of x. To specify the finite-visit convention, let

$$V_J = \{x \in \Sigma : \text{there are } 1 = M_1 < \cdots < M_J,\ M_i \equiv 1 \pmod 3,\ M_i x \in \Sigma \text{ for every } i\}.$$

Write

$$d_* := \frac{\log 2}{2 \log 3}, \qquad \Gamma_* := \lim_{J \to \infty} \dim_{\mathrm{H}} V_J.$$

The limit exists by monotonicity. This is the generalized nesting constant of [7, (1.14)]; a shift in the indexing convention for finite intersections does not change its limit. It is distinct from the dyadic constants $\Gamma_{\mathrm{H}}, \Gamma_{\mathrm{P}}$.

Theorem 15.1 (Exact generalized dimension)**.** *One has*

$$\boxed{\dim_{\mathrm{H}} E_* = \dim_{\mathrm{P}} E_* = \Gamma_* = \lim_{J \to \infty} \dim_{\mathrm{P}} V_J = d_*.}$$

Moreover,

$$\limsup_{J \to \infty} (\dim_{\mathrm{P}} V_J - d_*) \log \log J \le \frac{(\log 2)^2}{\log 3}. \tag{54}$$

The lower bound d_* was proved in [7, Theorems 2.4 and 6.3], using a construction of A. Bolshakov. We reproduce that construction below and do not claim its novelty. Our contribution here is the matching upper bound, the equality with the generalized nesting constant, the packing conclusions, and the quantitative estimate (54). The optimal rate of convergence is not asserted. Thus the result determines the dimension asked about in [7, Problem 1.6], beyond the positive lower bound already established there.

The proof uses a causal palette lemma specifically for pairs. We prove this finite lemma in full: no substitution of $r = 2$ into a container theorem requiring $r \ge 3$ is made. Throughout this section all logarithms are natural. Let Σ_q be the image of Σ in $\mathbb{Z}/3^q\mathbb{Z}$, represented in $[0, 3^q)$, and write $[z]_q$ for the canonical representative of z mod 3^q.

15.1 Capacity of adjacent translates

Lemma 15.2 (One-translate capacity)**.** *For $A, B \subset \mathbb{Z}/3^q\mathbb{Z}$, if $A + B \subset z + \Sigma_q$, then $|A||B| \le 2^q$.*

Proof. Empty sets cause no difficulty. Translate A to remove z and argue by induction on q, starting with the trivial case $q = 0$. The images of A, B modulo 3 have their sum in $\{0, 1\}$. Neither image can contain all three residues; if both contain two, their sum is all of $\mathbb{Z}/3\mathbb{Z}$, since two subsets of size two intersect after any translation. Hence one image is a singleton and the other has at most two elements. There are at most two nonempty pairs of least-digit sections.

For a pair of residues $a, b \in \{0, 1, 2\}$, write the corresponding sections as $a + 3A_a$ and $b + 3B_b$. Removing the least digit of their sum leaves $A_a + B_b$ in a single translate of Σ_{q-1}; the carry $\lfloor (a+b)/3 \rfloor$ is fixed. Thus induction bounds $|A_a||B_b|$ by 2^{q-1}. Summing over the at most two pairs proves the assertion. All these operations are in the indicated cyclic groups, so wraparound introduces no extra case. □

Lemma 15.3 (Sharp capacity for two adjacent translates)**.** *Let $q \geq 1$, $A, B \subset \Sigma_q$ and $u \in \mathbb{Z}_3^\times$. If*

$$uA + B \subset (z + \Sigma_q) \cup (z + 1 + \Sigma_q) \quad \text{in } \mathbb{Z}/3^q\mathbb{Z},$$

then $|A||B| \leq 2^{q+1}$. This bound is sharp.

Proof. Replace uA by $uA - z$. Each of $uA - z$ and B has at most two residues modulo 3. The exact identity

$$\Sigma_q \cup (1 + \Sigma_q) = \{0, 1, 2\} + 3\Sigma_{q-1}$$

shows that, for each of the at most four pairs of least-digit sections, the sum of the upper sections lies in a single translate of Σ_{q-1}. Lemma 15.2 bounds each product of section sizes by 2^{q-1}. Multiplication by u and translation preserve cardinalities, so their sum gives 2^{q+1}. Equality is attained by $u = 1$, $z = 0$, $A = \{0, 1\}$ and $B = \Sigma_q$. Adjacency of the two translates is essential to this argument. □

15.2 A causal palette lemma for pairs

The following statement permits overlapping windows. In particular, it does not assume that the letters associated with different positions are independent.

Lemma 15.4 (Causal pair palettes)**.** *Let W be a nonempty finite set of input words of length L. For $0 \leq j < N \leq L$, assign to each $x \in W$ a letter $Z_j(x) \in \Omega$, where $|\Omega| = Q$. Let a fixed simple directed graph have edges (j, k) with $j < k$. For each edge let $\mathcal{R}_{jk}(x_{<j}) \subset \Omega^2$ depend only on the input prefix preceding position j, and suppose that*

$$(Z_j(x), Z_k(x)) \in \mathcal{R}_{jk}(x_{<j}) \qquad (x \in W).$$

For an integer $T \geq 2Q$, put $\theta = Q/T$. Each $x \in W$ admits a deterministic fingerprint $F(x)$, consisting of at most $\lfloor \theta N \rfloor$ pairs $(j, Z_j(x))$, and nonempty palettes $A_j(x) \subset \Omega$ with the following properties:

(i) $Z_j(x) \in A_j(x)$, and $A_j(x)$ is determined by $(x_{<j}, F(x))$;
(ii) $A_j(x) \times A_k(x) \subset \mathcal{R}_{jk}(x_{<j})$ except on at most QTN edges;
(iii) for any probability distribution on W,

$$H(F) \leq N\{h(\theta) + \theta \log Q\}, \qquad h(t) = -t \log t - (1 - t) \log(1 - t).$$

Here H denotes Shannon entropy with natural logarithms. No locality assumption on the maps Z_j is required.

Proof. Initially give each position the mutable palette $S_j = \Omega$. Process source positions in increasing order. At source j, call $a \in S_j$ heavy if at least T outgoing edges (j, k) admit some $b \in S_k$ with $(a, b) \notin \mathcal{R}_{jk}(x_{<j})$.

If the actual letter $z_j = Z_j(x)$ is heavy, record (j, z_j) in F, set $A_j = \{z_j\}$, and remove from every future S_k all letters forbidden with source z_j on the edge (j, k), if that edge exists. These removals never delete the actual future letters. At least T distinct future letters are removed: the witnessing edges have distinct targets because the graph is simple. If z_j is light, let A_j be the set of all light letters in the current S_j, and make no removals. Once assigned, A_j is fixed; only palettes at unprocessed positions can shrink.

There are at most QN possible removals in total, so $|F|T \leq QN$. On every edge from a heavy source, the full product of the final palettes is allowed. At a light source, each of its at most Q assigned letters has fewer than T problematic edges. Subsequent shrinking cannot create new forbidden pairs. This gives at most QT bad edges per light source and at most QTN overall.

For replay at position j, use the known prefix $x_{<j}$ and the whole fingerprint F. At every earlier source $i \le j$, all required relations are known from the corresponding prefix. Source i is recorded exactly when its actual letter was heavy, in which case F supplies that letter. If it is not recorded, the algorithm needs only the computable set of light letters, not the unknown actual letter. Thus replay recovers A_j from the stipulated data.

Finally the number of possible fingerprints is at most

$$\sum_{m \le \lfloor \theta N \rfloor} \binom{N}{m} Q^m \le \exp\bigl(Nh(\theta) + \theta N \log Q\bigr).$$

For completeness, when $m \le \theta N$ and $\theta \le 1/2$, $\theta^m (1-\theta)^{N-m} \ge e^{-Nh(\theta)}$. Summing the corresponding terms in the binomial identity proves $\sum_{m \le \theta N} \binom{N}{m} \le e^{Nh(\theta)}$. The logarithm of the number of fingerprints bounds their entropy. □

15.3 Uniform affine counting from two windows

Fix distinct positive integers $E_1, \dots, E_K$ and units $u_i \in \mathbb{Z}_3^\times$, and put

$$c_i = 1 + 3^{E_i} u_i, \qquad R = \max_i E_i.$$

The shifts in the next lemma are arbitrary 3-adic integers. Their uniform treatment will also permit localization to every cylinder.

Lemma 15.5 (Finite affine bound)**.** *Let $q \ge 1$, $Q = 2^q$, $T \ge 2Q$ an integer, and $\theta = Q/T$. Define*

$$\sigma = d_* + \frac{d_*}{q} + \frac{h(\theta)}{\log 3} + \theta q \frac{\log 2}{\log 3} + \frac{QT}{K} \frac{\log 2}{\log 3}. \tag{55}$$

For $\beta_1, \dots, \beta_K \in \mathbb{Z}_3$, let

$$W_L(\beta) = \{X \in \Sigma_L : [c_i X + \beta_i]_L \in \Sigma_L \ (1 \le i \le K)\}.$$

For every $L \ge q$, with $N = L - q + 1$, a nonempty $W_L(\beta)$ satisfies

$$\log |W_L(\beta)| \le N\sigma \log 3 + (R + q - 1) \log 2. \tag{56}$$

In particular, whether or not it is empty,

$$|W_L(\beta)| \le 2^{R+q-1} 3^{\sigma L}. \tag{57}$$

These bounds are uniform in all the shifts β_i.

Proof. Assume $W_L(\beta)$ is nonempty. For $0 \le j < N$, use the window letter

$$Z_j(X) = B_{j,q}(X) := \left\lfloor X/3^j \right\rfloor \bmod 3^q \in \Sigma_q.$$

For each i include the edge $(j, v) = (j, j + E_i)$ whenever $v < N$. The distinctness of the E_i makes this a simple graph. For this edge set

$$P = [X]_j, \quad A = \beta_i + 3^{E_i} u_i P, \quad a_v = [A]_v, \quad D = \frac{A - a_v}{3^v} \in \mathbb{Z}_3.$$

Here D depends only on the fixed data and the prefix before j. An exact decomposition modulo 3^{v+q} gives

$$c_i X + \beta_i \equiv [X]_v + a_v + 3^v \bigl(B_{v,q}(X) + u_i B_{j,q}(X) + D\bigr).$$

Both $[X]_v$ and a_v are ordinary integers in $[0, 3^v)$. Their carry is therefore $e \in \{0, 1\}$, and admissibility implies

$$u_i B_{j,q}(X) + B_{v,q}(X) + D + e \in \Sigma_q \pmod{3^q} \quad \text{for some } e \in \{0, 1\}. \tag{58}$$

This argument is valid for arbitrary 3-adic u_i and β_i: only their canonical lower residues enter the ordinary carry.

Apply Lemma 15.4 with $\Omega = \Sigma_q$ and relations (58). On a good edge the full product of palettes lies in this relation. Its image under $(a, b) \mapsto u_i a + b$ lies in two adjacent translates of Σ_q, so Lemma 15.3 gives

$$w_j + w_v \le (q + 1) \log 2, \qquad w_j := \log |A_j| \le q \log 2.$$

On a bad edge use $w_j + w_v \le 2q \log 2$. The graph has at most KN edges, at most QTN of them bad. Compared with two incidences at each position for each shift, shift E_i loses at most $2 \min(E_i, N) \le 2R$ incidences. Consequently, for each input word,

$$\begin{aligned} 2K \sum_{j=0}^{N-1} w_j &\le KN(q + 1) \log 2 + 2q \log 2\, QTN + 2KRq \log 2, \\ \sum_{j=0}^{N-1} w_j &\le \frac{N(q + 1)}{2} \log 2 + q \log 2 \left(\frac{QTN}{K} + R \right). \end{aligned} \tag{59}$$

Using KN in the first term only enlarges the bound, including when some shifts exceed the window range.

Now choose X uniformly in $W_L(\beta)$ and write $h_t = H(X_t \mid X_{<t}, F)$. Replay in Lemma 15.4 ensures that, conditional on $(X_{<j}, F)$, the support of $B_{j,q}(X)$ lies in the corresponding palette. The entropy chain rule therefore gives

$$\sum_{t=j}^{j+q-1} h_t = H(B_{j,q}(X) \mid X_{<j}, F) \le \mathbb{E} w_j.$$

This identity, rather than independence of overlapping windows, is the reason their simultaneous use is legitimate. Each digit is covered at most q times; the total deficit from coverage q at every digit is $q(q - 1)$. Since $h_t \le \log 2$,

$$qH(X \mid F) \le \sum_j \mathbb{E} w_j + q(q - 1) \log 2.$$

The fingerprint is a deterministic function of X, whence $H(X) = H(F) + H(X \mid F)$. Substituting (59) and the fingerprint entropy bound gives (56), with precisely (55). Finally $N \le L$ and $\sigma \ge 0$ give (57); the empty case is immediate. □

15.4 Finite extraction and uniform local bounds

Lemma 15.6 (Distinct depths from a finite unit set)**.** *A set of more than $2 \cdot 3^{K-1}$ distinct elements of $\mathbb{Z}_3^\times$ contains an anchor λ_0 and $\lambda_1, \dots, \lambda_K$ such that $v_3(\lambda_i - \lambda_0)$ are distinct positive integers.*

Proof. One nonzero residue class modulo 3 contains more than 3^{K-1} elements. Form their tree of successive residues, stopping at a finite depth at which all elements are separated; such a depth exists because the set is finite and its elements are distinct. Suppress vertices with a single child, retaining the original depth at every branching vertex. The resulting tree has

at most three children per vertex. If every root-to-leaf path had at most $K-1$ branchings, induction would give at most 3^{K-1} leaves. Choose a path with at least K branchings and let its terminal leaf be λ_0. At each of K branchings choose a leaf in a different child subtree. The first differing positions of these leaves from λ_0 are exactly the distinct branching depths. They are positive because all leaves agree modulo 3. □

We use the usual Assouad dimension $\dim_{\mathrm{A}}$: an exponent a is admissible if every intersection of a set with a ball of radius ρ can be covered by at most $C(\rho/r)^a$ balls of radius r, with one C for all centers and $0 < r < \rho \le 1$.

Theorem 15.7 (Uniform finite intersections)**.** *For every $\varepsilon > 0$ there is an integer $J(\varepsilon)$ such that, for any $J \ge J(\varepsilon)$ distinct units $\lambda_1, \dots, \lambda_J \in \mathbb{Z}_3^\times$,*

$$\dim_{\mathrm{A}} \bigcap_{i=1}^{J} \lambda_i^{-1}\Sigma \le d_* + \varepsilon.$$

In particular the same bound holds for upper box, packing and Hausdorff dimensions. The covering constant may depend on the chosen multipliers; the exponent and sufficient cardinality do not.

Proof. Choose q, then T, then K so that the three nonconstant terms after d_* in (55) total less than ε: first $d_*/q < \varepsilon/3$, next

$$\frac{h(Q/T)}{\log 3} + \frac{Q}{T} q \frac{\log 2}{\log 3} < \frac{\varepsilon}{3},$$

and finally $(QT/K)(\log 2/\log 3) < \varepsilon/3$. Take $J(\varepsilon) > 2 \cdot 3^{K-1}$. By Lemma 15.6, choose an anchor and K companions. Multiplication by the anchor is an isometry; the image of the full intersection is contained in

$$G = \{y \in \Sigma : c_i y \in \Sigma \ (1 \le i \le K)\}, \qquad c_i = 1 + 3^{E_i} u_i,$$

with the distinct positive depths required in Lemma 15.5.

Consider any cylinder $P + 3^t\mathbb{Z}_3$ meeting G, with $0 \le P < 3^t$. Its lower input and output digits are already admissible. Writing $y = P + 3^t z$ yields the exact tail conditions

$$z \in \Sigma, \qquad c_i z + \beta_i \in \Sigma, \quad \beta_i = \frac{c_i P - [c_i P]_t}{3^t} \in \mathbb{Z}_3.$$

Thus the number of depth-$t+L$ cylinders meeting G inside it is at most $2^{R+q-1}3^{\sigma L}$ for $L \ge q$, uniformly in P, t. For $0 \le L < q$ the elementary bound $2^L \le 2^{R+q-1}$ suffices. This proves the local covering estimate on every pair of discrete 3-adic scales, with one constant. Rounding arbitrary radii to neighboring powers of 3 changes the constant only by a factor depending on σ. Hence $\dim_{\mathrm{A}} G \le \sigma < d_* + \varepsilon$. Monotonicity and the anchor isometry finish the proof. □

15.5 The exact value and its convergence rate

Proof of Theorem 15.1. For a sufficiently large fixed J, the set V_J is a countable union, over integer multiplier tuples, of the compact intersections in Theorem 15.7. Therefore $\dim_{\mathrm{P}} V_J \le d_* + \varepsilon$ by countable stability of packing dimension. No countable stability of upper box or Assouad dimension is used. Since $E_* \subset V_J$ for every J, this proves

$$\dim_{\mathrm{P}} E_* \le d_*, \qquad \Gamma_* \le d_*, \qquad \lim_{J\to\infty} \dim_{\mathrm{P}} V_J \le d_*.$$

The inclusion also covers zero; at a nonzero point the permissible unit multipliers are 1 (mod 3), and the multiplier 1 is available because the point lies in Σ.

For the known lower bound, consider

$$U = \left\{ \sum_{j \geq 0} \epsilon_j 3^{2j} : \epsilon_j \in \{0,1\} \right\}.$$

For every $k \geq 0$, the supports of $x \in U$ and $3^{2k+1}x$ have opposite parities, so $(1+3^{2k+1})x \in \Sigma$ without carry. Thus $U \subset E_* \cap V_J$ for every J. This is the Bolshakov construction used in [7, Theorem 6.3]. There are $2^{\lceil L/2 \rceil}$ depth-L cylinders of U. The fair Bernoulli measure on the even digits assigns each such cylinder mass $2^{-\lceil L/2 \rceil} \leq 3^{-d_* L}$. The cylinder cover and the mass distribution principle give $\dim_{\mathrm{H}} U = \dim_{\mathrm{P}} U = d_*$. This proves the asserted exact values.

For the rate, take, for all sufficiently large J,

$$K = \left\lfloor \log_3 \frac{J-1}{2} \right\rfloor + 1, \qquad q = \left\lfloor \frac{\log K - 5 \log\log K}{\log 4} \right\rfloor, \qquad Q = 2^q, \quad T = Qq^3.$$

Then $q \geq 2$, $\theta = q^{-3} \leq 1/2$, and $2 \cdot 3^{K-1} \leq J - 1 < J$, so finite extraction is available for every tuple in the definition of V_J. Moreover,

$$h(\theta) = O(q^{-3} \log q), \qquad \theta q = q^{-2}, \qquad \frac{QT}{K} = \frac{4^q q^3}{K} \leq \frac{q^3}{(\log K)^5} = O(q^{-2}).$$

Consequently (55) gives $\sigma = d_* + d_*/q + O(q^{-2})$. The same countable-union argument using (57) implies $\dim_{\mathrm{P}} V_J \leq \sigma$. Finally $q \sim \log K / \log 4$, $\log K \sim \log\log J$, and $d_* \log 4 = (\log 2)^2 / \log 3$. This proves (54). The constants involving R disappear in the dimension estimate for each fixed tuple before the countable union is taken. □

15.6 A sharp local bound for fixed infinite families

Theorem 15.8 (Fixed infinite intersections)**.** *For every infinite set of distinct units $\Lambda \subset \mathbb{Z}_3^\times$,*

$$F_\Lambda := \bigcap_{\lambda \in \Lambda} \lambda^{-1}\Sigma \quad \textit{satisfies} \quad \dim_{\mathrm{A}} F_\Lambda \leq d_*.$$

The bound is sharp. For $\Lambda_{\mathrm{odd}} = \{1\} \cup \{1 + 3^{2k+1} : k \geq 0\}$,

$$F_{\Lambda_{\mathrm{odd}}} = U \cup 3U, \qquad \dim_{\mathrm{A}} F_{\Lambda_{\mathrm{odd}}} = d_*.$$

Proof. For each $\varepsilon > 0$, choose any $J(\varepsilon)$ distinct members of Λ. The full intersection is contained in this finite intersection, so Theorem 15.7 and monotonicity give $\dim_{\mathrm{A}} F_\Lambda \leq d_* + \varepsilon$. One fixed subfamily for each exponent suffices; no uniform covering constant over all finite subfamilies, nor countability of Λ, is required.

In the displayed example, adding x and $3^h x$ never creates a carry, since each sum of digits is at most two. Their sum omits the digit two exactly when no two nonzero digits of x are distance h apart. Requiring this for every positive odd h forces all nonzero digits to have the same parity. This is precisely $x \in U \cup 3U$. In each cylinder of U, the number of descendants L levels below is at most $2^{\lceil L/2 \rceil}$; hence $\dim_{\mathrm{A}} U \leq d_*$. The reverse inequality follows from its Hausdorff dimension, and the stated sharpness follows. □

Remark 15.9 (The family must be fixed). Theorem 15.8 does not imply $\dim_{\mathrm{A}} E_* \leq d_*$. Every finite $0,1$-digit integer x belongs to E_*: for arbitrarily large h, the finite supports of x and

$3^h x$ are disjoint, so $(1+3^h)x \in \Sigma$. These integers are dense in Σ. Thus the upper box dimension of E_* is $\log_3 2$, and monotonicity together with $\overline{\dim}_{\mathrm{B}} E_* \le \dim_{\mathrm{A}} E_*$ gives

$$\overline{\dim}_{\mathrm{B}} E_* = \dim_{\mathrm{A}} E_* = \log_3 2.$$

For E_* the multiplier family can depend on the point; for F_Λ the same family works for every point. In particular $1 \in E_*$, and none of these generalized results excludes 1 from the dyadic exceptional set E_2.

16 Exact dimensions for Haar-typical multipliers

For a unit $\lambda \in \mathbb{Z}_3^\times$ put

$$C_\lambda = \Sigma \cap \lambda^{-1}\Sigma, \qquad H = 1 + 3\mathbb{Z}_3, \qquad S_\lambda = C_\lambda \cap H, \qquad b = \log_3(4/3).$$

Write m for normalized additive Haar measure on $\mathbb{Z}_3$ and m_H for its normalized restriction to H. Abram and Lagarias [7, Section 2.4] propose studying the dimension function $\lambda \mapsto \dim_{\mathrm{H}} C_\lambda$ for general 3-adic parameters, including its typical behavior. The next theorem determines its Haar-typical value on each of the two residue classes of units. It does not determine all exceptional parameters or the Baire-typical value.

Theorem 16.1 (Exact Haar-typical dimension)**.** *For m_H-almost every λ, simultaneously for $K = C_\lambda$ and $K = S_\lambda$,*

$$\dim_{\mathrm{H}} K = \dim_{\mathrm{P}} K = \underline{\dim}_{\mathrm{B}} K = \overline{\dim}_{\mathrm{B}} K = \log_3(4/3). \tag{60}$$

For every $\lambda \in 2 + 3\mathbb{Z}_3$ one has $C_\lambda = \{0\}$.

The proof computes an averaged second moment exactly, proves positivity of the resulting affine density, and transfers affine block entropy to the homogeneous intersection. The last transfer is separate from the almost-everywhere assertion about the affine translation: one cannot obtain it by setting that translation equal to zero. Throughout this section logarithms without a subscript are natural. For $z \in \mathbb{Z}_3$, let $[z]_n \in \{0, \dots, 3^n - 1\}$ be its canonical residue modulo 3^n, and let Σ_n be the 2^n canonical residues with digits $0, 1$.

16.1 Affine fibers and their exact second moment

For $\lambda \in H$ define

$$\begin{aligned} R_n(\lambda, \beta) &= \#\{x \in \Sigma_n : [\lambda x + \beta]_n \in \Sigma_n\}, \\ H_n(\lambda) &= 3^{-n} \sum_{\beta \bmod 3^n} \log R_n(\lambda, \beta). \end{aligned} \tag{61}$$

For every $\beta \in \mathbb{Z}_3$,

$$1 \le R_n(\lambda, \beta) \le 2^n, \qquad 3^{-n} \sum_{\beta \bmod 3^n} R_n(\lambda, \beta) = (4/3)^n. \tag{62}$$

Indeed, after an admissible input prefix the next output digit is $c + a$ modulo 3, where $a \in \{0, 1\}$. At least one of $c, c+1$ belongs to $\{0, 1\}$ modulo 3. This proves both nonemptiness and extension of every admissible finite prefix to an infinite point. For the average, each of the 4^n ordered pairs of input and output words determines a unique translation modulo 3^n.

Let μ be the probability measure on Σ with independent fair digits, and let ν_λ be the law of $Y - \lambda X$ for independent X, Y with law μ. Its cylinder density is

$$D_{\lambda,n}(\beta) = 3^n \nu_\lambda(\beta + 3^n\mathbb{Z}_3) = (3/4)^n R_n(\lambda, \beta). \tag{63}$$

Lemma 16.2 (Exact averaged second moment)**.** *For every $n \geq 1$,*

$$\int_H \|D_{\lambda,n}\|^2_{L^2(m)}\, dm_H(\lambda) = \frac{3}{2} - \frac{1}{2}\left(\frac{3}{4}\right)^n. \tag{64}$$

Consequently ν_λ has an $L^2(m)$ density for m_H-almost every λ.

Proof. Take independent X, X', Y, Y' with law μ, and set $A = X - X'$, $B = Y - Y'$. Counting coincident residue classes gives

$$\|D_{\lambda,n}\|_2^2 = 3^n \mathbb{P}(B \equiv \lambda A \pmod{3^n}).$$

The first differing digit of X, X' gives

$$\mathbb{P}(v_3(A) \geq n) = 2^{-n}, \qquad \mathbb{P}(v_3(A) = k) = 2^{-k-1} \quad (k \geq 0).$$

Conditioned on $v_3(A) = k$, the first unit digit of $A/3^k$ is $+1$ or -1 with equal probabilities; the analogous digit of $B/3^k$ is independent. If both valuations are at least n, the congruence holds. If both equal $k < n$, their leading signs agree with probability $1/2$. On this event the required residue of $\lambda \in H$ modulo 3^{n-k} has measure $3^{-(n-k-1)}$. Unequal valuations below n cannot contribute. Hence the average is

$$\begin{aligned} 3^n\left[4^{-n} + \sum_{k=0}^{n-1} 4^{-k-1}\frac{1}{2}\,3^{-(n-k-1)}\right] &= \left(\frac{3}{4}\right)^n + \frac{3}{8}\sum_{k=0}^{n-1}\left(\frac{3}{4}\right)^k \\ &= \frac{3}{2} - \frac{1}{2}\left(\frac{3}{4}\right)^n. \end{aligned}$$

For fixed λ, the functions $D_{\lambda,n}$ are a nonnegative martingale for the increasing cylinder sigma-fields of β under m. Their squared L^2 norms increase. Monotone convergence and (64) yield

$$\int_H \sup_n \|D_{\lambda,n}\|_2^2\, dm_H(\lambda) = \frac{3}{2}.$$

For almost every λ the martingale is therefore bounded in L^2 and converges in L^2 and almost everywhere to some D_λ. Its integrals on every cylinder agree with ν_λ; since cylinders generate the Borel sigma-field, $\nu_\lambda = D_\lambda m$. In particular $\int D_\lambda\, dm = 1$. □

Lemma 16.3 (Positivity of an affine density)**.** *Fix $\lambda \in H$. If $\nu_\lambda \ll m$, its density is positive m-almost everywhere.*

Proof. Decomposing the first input and output digits gives

$$\nu_\lambda = \frac{1}{4}\sum_{d \in (0,1,-\lambda,1-\lambda)} (z \mapsto d + 3z)_* \nu_\lambda,$$

where the four terms retain their multiplicities. A nonnegative version D of the density consequently satisfies, almost everywhere,

$$D(z) = \frac{3}{4}\sum_{\substack{d \in (0,1,-\lambda,1-\lambda) \\ d \equiv z \ (3)}} D((z-d)/3). \tag{65}$$

The three representatives $d_0 = 0, d_1 = 1, d_2 = -\lambda$ lie in different residue classes. Define $T(z) = (z - d_r)/3$ when $z \equiv r \pmod 3$. Each inverse branch scales Haar measure by $1/3$, so T preserves m. It is also ergodic. In fact,

$$(r_0, r_1, \ldots) \longmapsto \sum_{j \geq 0} 3^j d_{r_j}$$

is an isometric bijection from the full ternary sequence space onto $\mathbb{Z}_3$. Distinct representatives modulo 3 preserve the first differing position, and repeated application of T gives the inverse coding. The remainder $3^n T^n(z)$ tends to zero. Each sequence cylinder maps onto a residue class of the same measure, so this coding sends uniform Bernoulli measure to m and conjugates T to the ergodic one-sided shift.

For $Z = \{D = 0\}$, nonnegativity in (65) implies $Z \subseteq T^{-1}Z$ modulo null sets. The two sets have the same measure, hence agree modulo null sets. Ergodicity gives $m(Z) \in \{0, 1\}$, and $\int D\, dm = 1$ excludes $m(Z) = 1$. All uses of almost-everywhere identities are legitimate because the inverse branches identify Haar measure with its normalized restrictions to residue classes. □

Corollary 16.4 (Saturation of affine block entropy)**.** *For m_H-almost every λ,*

$$\lim_{n\to\infty} \frac{H_n(\lambda)}{n} = \log(4/3). \tag{66}$$

Proof. For almost every λ, Lemmas 16.2 and 16.3 give $0 < D_\lambda(\beta) < \infty$ for almost every β, with $D_{\lambda,n}(\beta) \to D_\lambda(\beta)$. Thus $n^{-1} \log D_{\lambda,n}(\beta) \to 0$. By (62) and (63),

$$-\log(4/3) \le \frac{1}{n} \log D_{\lambda,n}(\beta) \le \log(3/2).$$

Bounded convergence now gives

$$\frac{H_n(\lambda)}{n} = \log(4/3) + \int \frac{1}{n} \log D_{\lambda,n}\, dm \longrightarrow \log(4/3).$$

No integrability assumption on $\log D_\lambda$ is needed. □

16.2 Transfer to the homogeneous fiber

Lemma 16.5 (Block measures on a prescribed homogeneous fiber)**.** *For each fixed integer $q \ge 1$ and m_H-almost every λ, there is an exact-dimensional probability measure on S_λ of dimension $H_q(\lambda)/(q \log 3)$. In particular,*

$$\dim_{\mathrm{H}} S_\lambda \ge \frac{H_q(\lambda)}{q \log 3}. \tag{67}$$

Proof. Partition H into its finitely many residue classes modulo 3^q. In one such class fix $\ell = [\lambda]_q$. This also fixes $H_q(\lambda) = H_q(\ell)$. Choose an admissible unit prefix $P_q \in \Sigma_q$ deterministically from ℓ: start with digit 1 and take the least admissible digit thereafter. The extension property proved after (62) ensures this choice exists.

Suppose an admissible unit prefix P_j of length $j = tq$ has been chosen. An input block $z \in \Sigma_q$ produces the output block

$$[\ell z + \beta_j]_q, \qquad \beta_j = \left[\frac{\lambda P_j - [\lambda P_j]_j}{3^j}\right]_q. \tag{68}$$

There are exactly $R_q(\ell, \beta_j)$ admissible blocks. Give them equal probabilities and set $P_{j+q} = P_j + 3^j z$. For each fixed λ these consistent cylinder probabilities define a probability measure $\eta_{\lambda,q}$ supported on S_λ. They depend on finitely many parameter digits at each finite stage, so $\lambda \mapsto \eta_{\lambda,q}$ is a measurable probability kernel.

To analyze the measure, first choose λ with conditional Haar law in the class ℓ, then implement the block choices using independent uniform random variables $V_1, V_2, \ldots$. Before the choice at $j = tq$, the past sigma-field is

$$\mathcal{F}_t = \sigma([\lambda]_{tq}, V_1, \ldots, V_{t-1}).$$

The prefix P_j is measurable with respect to this past. The fresh parameter block $L_j = [(\lambda - [\lambda]_j)/3^j]_q$ is uniform modulo 3^q and independent of $\mathcal{F}_t$. Equation (68) gives

$$\beta_j \equiv c_j + L_j P_j \pmod{3^q}, \tag{69}$$

where c_j is determined by the past. Since $P_j \equiv 1 \pmod 3$, multiplication by P_j permutes all residues modulo 3^q. Thus β_j has the uniform conditional law for every past. After using V_t, the next prefix is measurable with respect to $\mathcal{F}_{t+1}$, which proves the induction.

Consequently $Y_t = \log R_q(\ell, \beta_{tq})$ are independent and identically distributed in this joint experiment, with $0 \le Y_t \le q \log 2$ and expectation $H_q(\ell)$. At the block boundary $n = q(T+1)$ the mass of the actual cylinder is exactly

$$\eta_{\lambda,q}(x + 3^n \mathbb{Z}_3) = \prod_{t=1}^{T} R_q(\ell, \beta_{tq})^{-1}.$$

The strong law gives local dimension $H_q(\ell)/(q \log 3)$ along these boundaries, jointly almost surely. Conditional on the full parameter λ, the V_t still implement exactly $\eta_{\lambda,q}$. Fubini's theorem therefore gives the same almost-sure local dimension for almost every fixed λ. The parameter marginal remains Haar throughout; no independence of the shifts for a fixed λ is asserted.

Between consecutive block boundaries a cylinder mass is bounded by the two boundary masses. Their ratio is at most 2^q, and the depths differ by q. Dividing logarithms by $n \log 3$ shows that the local dimension exists on all scales. Exact dimensionality implies (67): for any smaller exponent, restrict the measure to a positive-measure set on which the corresponding cylinder upper bound holds beyond one common depth, and apply the mass distribution principle. Taking the union of the finitely many residue classes completes the proof. □

Intersect the full-measure parameter sets from Lemma 16.5 for every $q \ge 1$, and also the full-measure set of Corollary 16.4. On their intersection,

$$\dim_{\mathrm{H}} S_\lambda \ge \sup_{q \ge 1} \frac{H_q(\lambda)}{q \log 3} \ge b. \tag{70}$$

This is the required passage to translation zero. It uses fresh parameter blocks rather than a substitution into an almost-everywhere statement in the translation variable.

16.3 The upper bound and the initial zero digits

Let $Q_n(\lambda)$ be the number of admissible unit prefixes of length n. Every such prefix extends, so this is the exact number of depth-n cylinders meeting S_λ. For each of the 2^{n-1} possible unit input words P, the residue λP is uniform among the 3^{n-1} residues congruent to 1 modulo 3 when λ has law m_H. Exactly 2^{n-1} of those residues have digits $0, 1$. Hence

$$\mathbb{E}_H Q_n = (4/3)^{n-1}. \tag{71}$$

For every $\varepsilon > 0$, Markov's inequality gives

$$m_H\{Q_n > 3^{n(b+\varepsilon)}\} \le \tfrac{3}{4}\, 3^{-n\varepsilon}.$$

This is summable. Borel–Cantelli, followed by a countable intersection over positive rational ε, shows that almost surely $\overline{\dim}_{\mathrm{B}} S_\lambda \le b$.

To handle C_λ, count the possible position of the first digit 1. With $N_n(C_\lambda)$ denoting its number of depth-n cylinders,

$$N_n(C_\lambda) = 1 + \sum_{j=1}^{n} Q_j(\lambda). \tag{72}$$

The first term is the all-zero prefix. A prefix with $v < n$ initial zeros corresponds to a unit prefix of length $n - v$. For almost every λ and every $\varepsilon > 0$, the finitely many exceptions can be absorbed into a finite constant: $Q_j(\lambda) \le K_{\lambda,\varepsilon} 3^{j(b+\varepsilon)}$ for all j. A geometric sum in (72) therefore gives $\overline{\dim}_{\mathrm{B}} C_\lambda \le b$. This argument does not use countable stability of box dimension. Together with (70), the inclusion $S_\lambda \subseteq C_\lambda$, and the inequalities

$$\dim_{\mathrm{H}} \le \dim_{\mathrm{P}} \le \overline{\dim}_{\mathrm{B}}, \qquad \dim_{\mathrm{H}} \le \underline{\dim}_{\mathrm{B}} \le \overline{\dim}_{\mathrm{B}},$$

it proves (60). Finally, if $\lambda \equiv 2 \pmod 3$ and $x \in \Sigma \setminus \{0\}$, write $x = 3^v u$ with $u \equiv 1 \pmod 3$. The first nonzero digit of λx is 2, proving the remaining assertion of Theorem 16.1.

Corollary 16.6 (The canonical measure has smaller dimension)**.** *On S_λ, start with digit 1, choose each of the two admissible digits with probability $1/2$ at every free step, and take the forced digit at every other step. For almost every $\lambda \in H$ this probability measure has full support S_λ and is exact-dimensional of dimension $(\log_3 2)/3$, strictly less than $\dim_{\mathrm{H}} S_\lambda = \log_3(4/3)$.*

Proof. This is the measure in Lemma 16.5 with $q = 1$. The initial prefix is 1, and every subsequent admissible cylinder has positive mass, so the support is all of S_λ. For the three translations modulo 3, the counts R_1 are $2, 1, 1$. Thus $H_1 = (\log 2)/3$, giving the stated dimension. The strict inequality follows from $(4/3)^3 > 2$. □

16.4 All unit parameters: a sharp gap and oscillation

The almost-everywhere theorem leaves individual parameters unclassified. A complementary uniform bound follows by extending the digit coding of Abram, Bolshakov and Lagarias [2, Theorem 6.2] from integer multipliers to arbitrary 3-adic units. The extension below is isometric and controls cylinder counts uniformly, giving an Assouad bound.

Theorem 16.7 (Isometric coding and sharp dimension gap)**.** *Put $\varphi = (1 + \sqrt{5})/2$ and $g = \log_3 \varphi$. If $\lambda \in H \setminus \{1\}$ and $k = v_3(\lambda - 1)$, then C_λ admits an isometric embedding into*

$$\mathcal{H}_k = \left\{ \sum_{j \ge 0} b_j 3^j : b_j \in \{0,1\},\ b_j b_{j+k} = 0 \text{ for every } j \ge 0 \right\}.$$

Consequently $\dim_{\mathrm{A}} C_\lambda \le g$. The bound is sharp: $C_{1+3^k} = \mathcal{H}_k$ has Hausdorff, packing, lower and upper box, and Assouad dimensions g for every $k \ge 1$. For $\lambda = 1 - 3^k$, all these dimensions are zero.

Proof. Write $\lambda = 1 + 3^k u$ with $u \in \mathbb{Z}_3^\times$. For an admissible input prefix $P_j = \sum_{i<j} a_i 3^i$ set

$$r_j = \frac{\lambda P_j - [\lambda P_j]_j}{3^j}, \qquad d_j = [r_j]_1.$$

If $d_j = 0$ both input digits are allowed; if $d_j = 1$ only 0 is allowed; if $d_j = 2$ only 1 is allowed. Define, for $r \in \mathbb{Z}_3$,

$$F(r) = \frac{r - [r]_1}{3} + \mathbf{1}_{\{[r]_1 = 2\}}.$$

Every admissible step satisfies

$$r_{j+1} = F(r_j) + 3^{k-1}ua_j, \qquad F(r+3^v t) = F(r) + 3^{v-1}t \quad (v \geq 1). \tag{73}$$

At a free position j and for either alternative $a \in \{0,1\}$, put

$$\rho_j(a) = \left[F^{k-1}(F(r_j) + 3^{k-1}ua)\right]_1.$$

This is the residue of r_{j+k} for every admissible choice of the intervening digits. Indeed, a digit inserted strictly after j contributes $3^{k-1}u$ times that digit and undergoes fewer than $k-1$ further applications of F before that residue is read. By (73) its contribution is still a multiple of 3. Comparing the trajectories one insertion at a time proves the assertion, including $k=1$ where there are no intervening digits. The two choices at j instead give

$$\rho_j(1) - \rho_j(0) \equiv u \not\equiv 0 \pmod 3.$$

Thus at most one alternative makes position $j+k$ free.

Encode a forced position by $b_j = 0$. At a free position, if exactly one alternative has $\rho_j(a) = 0$, encode that alternative by 0 and the other by 1; if neither has zero residue, put $b_j = a_j$. Whenever $b_j = 1$, position $j+k$ is forced, so $b_{j+k} = 0$. The code belongs to $\mathcal{H}_k$ and depends only on the input prefix through j. If two inputs first differ at j, their common past makes position j free, and its two choices receive different code digits. Their first differing position is preserved exactly, which proves isometry.

Within L consecutive positions the constraint defining $\mathcal{H}_k$ separates into k residue classes. If class r occurs ℓ_r times, it has F_{ℓ_r+2} words without adjacent ones, where $F_0 = 0, F_1 = 1$ are the Fibonacci numbers. The same product is an upper bound after any fixed prefix, since boundary constraints can only remove possibilities. As $F_{\ell+2} \leq \varphi^{\ell+1}$, the number of continuations is at most φ^{L+k}. This uniform cylinder estimate gives $\dim_{\mathrm{A}} \mathcal{H}_k \leq g$. There is no carry in $x + 3^k x$ for $x \in \Sigma$, so the condition that it belong to Σ is exactly $a_j a_{j+k} = 0$. Hence $C_{1+3^k} = \mathcal{H}_k$. Its Hausdorff dimension g is the known family computation of [7, Theorems 2.3 and 5.5]; combined with the Assouad upper bound, this gives all the stated dimension equalities.

For $\lambda = 1 - 3^k$, subtract $3^k x$ digit by digit from x. If a first borrow occurs, its output digit is 2, which is forbidden. Thus admissibility is equivalent to $a_j \geq a_{j-k}$ for every $j \geq k$. Along each residue class modulo k the sequence is $0^t 1^\infty$, with $t \geq 0$, or 0^∞. This gives a countable set. More strongly, after any fixed prefix the number of length-L continuations is at most $(L+1)^k$, which is bounded by $C_{\varepsilon,k} 3^{\varepsilon L}$ for every $\varepsilon > 0$. Its Assouad dimension, and hence all the listed dimensions, are zero. □

Corollary 16.8 (Exact punctured limits at the identity)**.** *For $f(\lambda) = \dim_{\mathrm{H}} C_\lambda$, in the 3-adic topology,*

$$\liminf_{\substack{\lambda \to 1 \\ \lambda \neq 1}} f(\lambda) = 0, \qquad \limsup_{\substack{\lambda \to 1 \\ \lambda \neq 1}} f(\lambda) = \log_3 \varphi < \log_3 2 = f(1).$$

The same conclusions hold with packing, either box dimension, or Assouad dimension in place of Hausdorff dimension. No unit parameter has a dimension in $(\log_3 \varphi, \log_3 2)$, and only $\lambda = 1$ has dimension $\log_3 2$.

Proof. The sequences $1 - 3^k$ and $1 + 3^k$ tend to 1 and attain the two limiting values, while Theorem 16.7 provides the uniform upper bound away from 1 in H. Parameters in $2 + 3\mathbb{Z}_3$ have intersection $\{0\}$, and $C_1 = \Sigma$ has all the stated dimensions $\log_3 2$. □

Remark 16.9 (Scope of the parameter theorems)*.* Theorem 16.1 concerns almost every multiplier in an uncountable parameter space. It gives no conclusion about a prescribed integer multiplier or the sequence 4^n. Theorem 16.7 is uniform but is an upper bound, not a classification of the spectrum below $\log_3 \varphi$. Neither statement decides $1 \notin E_2$.

17 Fractional parts and fixed moving Cantor targets

The full real expansion condition in Theorem 7.1 restricts digits on both sides of the ternary point. A different question, posed by Lagarias in [5, Section 5], restricts only the fractional part. Put

$$K_{01} = \left\{ \sum_{j\geq 1} \varepsilon_j 3^{-j} : \varepsilon_j \in \{0,1\} \right\} \subset [0,1/2], \qquad \mathcal{C}_{\mathrm{per}} = \mathbb{Z} + K_{01}.$$

The set $\mathcal{C}_{\mathrm{per}}$ is closed, since this union of integer translates is locally finite. On $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, define

$$E_{\mathbb{T}} = \{x \in \mathbb{T} : \{2^n x\} \in K_{01} \text{ for infinitely many } n \geq 0\}.$$

Here $\{y\} = y - \lfloor y \rfloor$; the condition is independent of the representative of x. For the standard middle-thirds set $K_{02} = 2K_{01}$, also put

$$E_{\mathrm{circ}} = \{x \in \mathbb{T} : \{2^n x\} \in K_{02} \text{ for infinitely many } n \geq 0\}.$$

The set $E_{\mathbb{T}}$ is the fractional-part exceptional set in Lagarias's question; E_{circ} is its additional standard middle-thirds variant. We answer both dimension questions and prove the following stability under translations. The empty set is assigned Hausdorff and packing dimension zero.

Theorem 17.1 (Fixed moving targets)**.** *For every fixed sequence $\boldsymbol{\tau} = (\tau_n)_{n\geq 0}$ of real numbers, the set*

$$E_{\mathbb{T}}(\boldsymbol{\tau}) = \{x \in [0,1] : 2^n x + \tau_n \in \mathcal{C}_{\mathrm{per}} \text{ for infinitely many } n \geq 0\}$$

has Hausdorff and packing dimension zero. The shifts may be irrational and unbounded. Similarly, for every fixed sequence $\boldsymbol{\xi} = (\xi_n)_{n\geq 0}$ in $\mathbb{Z}_3$, the set

$$E_3(\boldsymbol{\xi}) = \{x \in \mathbb{Z}_3 : 2^n x + \xi_n \in \Sigma \text{ for infinitely many } n \geq 0\}$$

has Hausdorff and packing dimension zero.

Corollary 17.2 (The circle question)**.** *The two fractional-part exceptional sets satisfy*

$$\dim_{\mathrm{H}} E_{\mathbb{T}} = \dim_{\mathrm{P}} E_{\mathbb{T}} = \dim_{\mathrm{H}} E_{\mathrm{circ}} = \dim_{\mathrm{P}} E_{\mathrm{circ}} = 0.$$

The sequence of targets is fixed before the point is chosen. There is no simultaneous assertion for the union over all target sequences: choosing $\tau_n = -2^n x$ would include any prescribed real point. The new ingredient below is the transfer from a uniform affine finite-word estimate to a periodic real target. Its two arithmetic inputs, Lemma 8.1 and Proposition 9.2, were proved earlier in this manuscript.

17.1 The precise finite input

For every $\varepsilon > 0$, those inputs provide fixed finite extraction parameters with the following property. From any infinite $S \subseteq \mathbb{Z}_{\geq 0}$ one can choose distinct true indices $n_0, n_1, \ldots, n_m \in S$ and separated template data such that, for $c_i = 2^{n_i - n_0}$,

$$|W_L(\boldsymbol{\beta})| \leq C_{\mathcal{D}} 3^{\varepsilon L} \qquad (L \geq q,\ \boldsymbol{\beta} \in \mathbb{Z}_3^m), \tag{74}$$

where

$$W_L(\boldsymbol{\beta}) = \{U \in \Sigma_L : [c_i U + \beta_i]_L \in \Sigma_L \ (1 \leq i \leq m)\}.$$

The constant depends on the chosen finite template data $\mathcal{D}$, but not on L or the shifts. All admissible finite words are counted; infinite extendibility is not required.

For clarity, the parameters can be fixed before S is specified. In (28), choose k and then r so that $1/k + 2^{k+1}\log 2/(rk\log 3) < \varepsilon$, with a positive margin. Choose b with $3^b > r-1$ and put $q = b + k$. Using the thresholds

$$T_j = u^{2^{j-1}-1}\ (2 \le j \le r-1), \qquad T_{\mathrm{s}} = u^{2^{r-1}-1}, \qquad m = u^{2^r},$$

and then fixing a sufficiently large integer u, makes θ, ϵ_F, η in (26)–(28) small enough that $\theta \le 1/2$ and $s \le \varepsilon$. Apply Lemma 8.1 with $K = m$ and then Proposition 9.2; its constant is $C_{\mathcal{D}} = 3^{R+b+k-1}$. Some slopes c_i may be smaller than one. They are always positive rational 3-adic units.

17.2 An affine transfer to the periodic target

Lemma 17.3 (Periodic-target transfer)**.** *Let $c_i = p_i/q_i > 0$ be reduced rationals with $3 \nmid p_i q_i$. Suppose* (74) *holds for these slopes, uniformly in all shifts. For arbitrary fixed $d_1, \dots, d_m \in \mathbb{R}$ and a bounded closed interval I, the compact set*

$$F = \{Y \in I : Y \in \mathcal{C}_{\mathrm{per}}, \quad c_i Y + d_i \in \mathcal{C}_{\mathrm{per}}\ (1 \le i \le m)\}$$

has upper box dimension at most ε.

Proof. Compactness follows from closedness of the targets. Fix $L \ge q$ and $Y \in F$, and write

$$\begin{aligned} T &= \lfloor 3^L Y \rfloor, & V_i &= \lfloor 3^L (c_i Y + d_i) \rfloor, & D_i &= \lfloor 3^L d_i \rfloor, \\ t &= \{3^L Y\}, & t_i &= \{3^L (c_i Y + d_i)\}, & \theta_i &= \{3^L d_i\}. \end{aligned}$$

Let U be the representative of T modulo 3^L in $[0, 3^L)$. The target conditions imply $U \in \Sigma_L$ and $V_i \bmod 3^L \in \Sigma_L$. Indeed, for $z \in K_{01}$ the remainder after its first L ternary digits is between zero and $3^{-L}/2$, so the floor reads exactly that prefix, including terminating expansions. Negative Y causes no change, since $T = 3^L \lfloor Y \rfloor + \lfloor 3^L \{Y\} \rfloor$.

The exact rounding error is

$$\delta_i = V_i - c_i T - D_i = c_i t + \theta_i - t_i. \tag{75}$$

Its first expression gives $\delta_i \in q_i^{-1}\mathbb{Z}$ and its second gives $-1 < \delta_i < c_i + 1$. Thus

$$\delta_i \in \mathscr{D}_i = q_i^{-1}\mathbb{Z} \cap (-1, c_i + 1), \qquad |\mathscr{D}_i| = p_i + 2q_i - 1.$$

Every element of $\mathscr{D}_i$ is 3-adically integral. Set $\beta_i = D_i + \delta_i \in \mathbb{Z}_3$. Since $3^L \mid T - U$ in $\mathbb{Z}_3$,

$$c_i U + \beta_i = V_i - c_i (T - U) \equiv V_i \pmod{3^L}. \tag{76}$$

Consequently $U \in W_L(\boldsymbol{\beta})$.

At each fixed depth L, all D_i are fixed, while the vector of rounding errors has at most $\prod_i (p_i + 2q_i - 1)$ possibilities. If $I = [a, b]$, the integer part of Y has at most $J_I = \lfloor b \rfloor - \lfloor a \rfloor + 1$ possibilities. Each integer part and each U specify one interval of length 3^{-L} containing the corresponding points. Replacing these intervals by their closures gives a cover of F by at most

$$J_I \prod_{i=1}^{m} (p_i + 2q_i - 1)\, C_{\mathcal{D}} 3^{\varepsilon L} \tag{77}$$

such intervals. The shifts β_i may vary with L, and the d_i may be irrational: this is precisely why the uniformity in (74) is needed. No common shift vector across depths has been assumed. All factors before $3^{\varepsilon L}$ in (77) are independent of L. The consecutive scales 3^{-L} have bounded ratio, proving the claim. □

17.3 Countable compact covers

Proof of Theorem 17.1. Fix $\varepsilon > 0$ and the parameters of (74). If $x \in E_{\mathbb{T}}(\boldsymbol{\tau})$, its set of true visit times is infinite. Choose the extracted times from this set and put

$$Y = 2^{n_0}x + \tau_{n_0}, \qquad c_i = 2^{n_i - n_0}, \qquad d_i = \tau_{n_i} - c_i\tau_{n_0}.$$

Then Y and all $c_iY + d_i$ belong to $\mathcal{C}_{\text{per},}$ and $Y \in I = [\tau_{n_0}, \tau_{n_0} + 2^{n_0}]$. Lemma 17.3 gives a compact set F containing Y with $\overline{\dim}_{\mathrm{B}}F \le \varepsilon$. Its image under the inverse affine map $Y \mapsto 2^{-n_0}(Y - \tau_{n_0})$ is compact with the same bound.

There are countably many finite choices of visit indices, integer template coefficients and integer exponents. The sequence $\boldsymbol{\tau}$ is fixed, so I and the real shifts d_i are determined by these indices. The unit remainders in the templates are likewise determined by the slopes, integer coefficients and exponents. Hence these affine images form a countable compact cover of $E_{\mathbb{T}}(\boldsymbol{\tau})$, each member having upper box dimension at most ε. Countable stability of packing dimension yields

$$\dim_{\mathrm{H}} E_{\mathbb{T}}(\boldsymbol{\tau}) \le \dim_{\mathrm{P}} E_{\mathbb{T}}(\boldsymbol{\tau}) \le \varepsilon.$$

For the 3-adic statement, extract the same type of finite data from the true visit set of $x \in E_3(\boldsymbol{\xi})$. This time set

$$Y = 2^{n_0}x + \xi_{n_0}, \qquad \beta_i = \xi_{n_i} - c_i\xi_{n_0} \in \mathbb{Z}_3.$$

The compact set

$$G = \{Y \in \Sigma : c_iY + \beta_i \in \Sigma \ (1 \le i \le m)\}$$

has at most $|W_L(\boldsymbol{\beta})|$ prefix cylinders at depth L; therefore $\overline{\dim}_{\mathrm{B}}G \le \varepsilon$. The map $x \mapsto 2^{n_0}x + \xi_{n_0}$ is an isometry of $\mathbb{Z}_3$. Countably many inverse images, indexed by the finite integer data, cover $E_3(\boldsymbol{\xi})$. Its Hausdorff and packing dimensions are at most ε as well. Letting $\varepsilon \downarrow 0$ proves both assertions. □

Proof of Corollary 17.2. Take $\tau_n = 0$ for all n and project $[0,1]$ onto $\mathbb{T}$. This Lipschitz projection maps the exceptional set onto $E_{\mathbb{T}}$, so both dimensions are zero.

For the standard target, choose $x \in [0,1)$ representing a point of E_{circ}, and put $y = x/2$. At a visiting time write $2^n x = a + 2z$, with $a \in \mathbb{Z}$ and $z \in K_{01}$. Then $2^n y = a/2 + z$ belongs either to $\mathbb{Z} + K_{01}$ or to $\mathbb{Z} + 1/2 + K_{01}$ according to the parity of a. One of these two alternatives occurs infinitely often. Theorem 17.1, with the fixed shifts 0 and $-1/2$, places y in a union of two zero-dimensional exceptional sets. The similarity $y \mapsto 2y$ and the circle projection give the result for E_{circ}. The endpoint $1 \in K_{02}$ is identified with 0 on the circle; this makes no difference to the conclusion. □

Corollary 17.4 (Finitely many fixed targets per time)**.** *For a fixed positive integer a, allow at time n any of the fixed shifts $\tau_{n,1}, \dots, \tau_{n,a}$, or, in the 3-adic case, $\xi_{n,1}, \dots, \xi_{n,a}$. The set of points visiting one of the corresponding targets infinitely often still has zero Hausdorff and packing dimension.*

Proof. For each point with infinitely many visits, at least one of the a target indices occurs infinitely often. The exceptional set is therefore the union of the a exceptional sets in Theorem 17.1. □

Remark 17.5 (Scope and the distinction between dimensions). Every dyadic rational point of $\mathbb{T}$ belongs to both $E_{\mathbb{T}}$ and E_{circ}, because its orbit is eventually zero. Consequently $E_{\mathbb{T}}$ and E_{circ} are dense and have both lower and upper box dimension one, even though their packing dimensions are zero. The proof above uses a countable compact cover; it does not

assert a global subpower covering count. In particular, the class of 1 equals 0 on the circle and belongs to both exceptional sets. This circle statement does not exclude the ordinary starting value 1 from the 3-adic set E_2. Neither countability of the exceptional sets nor a finite-visit Assouad theorem on the circle is established here. The moving-target quantifiers do not include a union over all sequences or an unbounded number of alternatives at each time.

Verification added in version 4

Five additional finite audit programs check the generalized pair-capacity and carry identities, the causal palette construction and finite tree extraction, the all-unit isometric coding, the exact averaged second moment and block-shift permutations, and the circle transfer identities. Their exact scopes are recorded with the reproducible outputs. They supplement the proofs at arbitrary depths and do not infer infinite dimension assertions from finite samples. The assembled new English sections receive separate internal mathematical and integration reviews; the exact source hashes are recorded in the release. The earlier sections are preserved, not described as freshly reproved.

Verification added in version 3

The new standard-library Python checks verify finite digital fibers, all coefficient and target cases in the stated test ranges, complete prefix-cube identities, and finite carry, rational-denominator and sieve identities. These computations supplement the proofs for arbitrary depths and parameters; they do not prove an asymptotic by extrapolation. The exact new source files received fresh independent internal AI-assisted reviews. The earlier proof sections and causal-container appendix are retained from the released version 2 sources; this update does not describe their preservation as a fresh line-by-line audit. The metric branch uses no external unproved hypothesis. The counting branch explicitly states its external analytic input, Kurlberg–Pomerance Lemma 10 [8]; no GRH assumption is used.

Dependency and verification record

For the dimension proofs retained from version 2, the imported nonstandard finite result is Theorem 5.1, from Appendix A of the exact release v1.0.0 of [4], commit

`81c7f5ae45c3ce507173b82baca48fdfeb52c6ef`

Its complete proof is included unchanged in Appendix A. The SHA-256 digest of the appendix source `causal_container.tex` is

`1ae15961cd3bc705e7313c9e56de34e5127`
`7cffaa7cc3a58c6e4c9cb6e672750`

with the two displayed lines concatenated. This identifies the exact imported proof on which the finite entropy estimate depends. The unit-equation theorem is the published result [3]; finite Ramsey theory is used only in its ordinary finite form. The real argument also uses Kneser's addition theorem [6], through the full proof of Lemma 9.1 reproduced from [4, Lemma 2.1]. The statements and proofs of the uniform finite extraction and of the real infinite-set extraction are kept separate. The retained version-2 argument does not itself assert a real finite-visit nesting theorem or a finite-visit Assouad theorem. The fixed-target extensions proved in Section 17 have exactly the quantifiers stated there; no simultaneous theorem for arbitrary moving targets depending on the starting point is claimed.

The arithmetic extraction and the entropy transfer were subjected to independent internal AI-assisted reviews. Exact finite checks were also used to test determinant identities and low-dimensional cases. These checks do not replace the arguments for arbitrary d, all Ramsey refinements, or the causal-container theorem. AI assistance was used in developing proofs, writing the manuscript, and implementing checks. This is not external peer review or proof-assistant verification. Authorship is attributed solely to Timofei Iuzhakov.

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A The finite causal-container theorem and its proof

The following source is reproduced unchanged from Appendix A of [4], release v1.0.0. Its symbols are local to this appendix.

A.1 Exact theorem

Let $r \geq 3$, let $V_0 = \{0, \dots, N-1\}$, let Ω be a finite alphabet of size Q, and let $\mathcal{W} \subseteq \mathcal{X}^L$ be nonempty, where $\mathcal{X}$ is any finite input alphabet. For every $i \in V_0$, a word x determines a letter $Z_i(x) \in \Omega$. No locality hypothesis on the maps Z_i is needed for this finite theorem.

Let $\mathcal{E}$ be a fixed family of ordered positional edges

$$e = (s = i_0 < i_1 < \cdots < i_{r-1}).$$

For each edge there is a relation $\mathcal{R}_e(x_{<s}) \subseteq \Omega^r$, depending only on the indicated input prefix, and every $x \in \mathcal{W}$ satisfies

$$(Z_{i_0}(x), \ldots, Z_{i_{r-1}}(x)) \in \mathcal{R}_e(x_{<s}).$$

Assume:

1. for a fixed source s, the future position sets $e \setminus \{s\}$ of distinct outgoing edges are disjoint;
2. every unordered pair of positions belongs to at most C positional edges.

Fix positive integer thresholds $T_{\mathrm{s}}, T_2, \ldots, T_{r-1}$, and put

$$D = CQ^{r-3},$$

$$\theta = Q\left[\frac{D\sum_{j=2}^{r-1} T_j + 1}{T_{\mathrm{s}}} + \frac{1}{T_2} + D\sum_{j=3}^{r-1} \frac{1 + \sum_{l=2}^{j-1} T_l}{T_j}\right].$$

Assume $\theta \le 1/2$. Then each word can be assigned a set $F(x)$ of at most θN pairs $(i, Z_i(x))$, with at most one pair for each position, and nonempty sets $A_i(x) \subseteq \Omega$, such that:

- $Z_i(x) \in A_i(x)$;
- $A_i(x)$ is determined by $(x_{<i}, F(x))$;
- except for at most

$$Q\left(T_{\mathrm{s}} + D\sum_{j=2}^{r-1} T_j\right) N$$

edges, the full product of the assigned alphabets is compatible:

$$\prod_{\nu=0}^{r-1} A_{i_\nu}(x) \subseteq \mathcal{R}_e(x_{<i_0});$$

- for every distribution on $\mathcal{W}$,

$$H(F) \le N\big[h(\theta) + \theta \log Q\big], \qquad h(t) = -t\log t - (1-t)\log(1-t).$$

The window parameter q in the application only sets $N = L - q + 1$; the abstract theorem itself does not involve windows.

A.2 Literals, truth, and complete origins

Fix one realized word x for the construction and counting arguments, and write $z_i = Z_i(x)$. A **literal** is a pair (i, a) with $a \in \Omega$. A literal is true if $a = z_i$. There are QN literals.

A **complete origin** consists of:

- one positional edge $e = (s, i_1, \ldots, i_{r-1})$;
- one assignment $(a_1, \ldots, a_{r-1})$ of letters to its future positions;
- the condition

$$(z_s, a_1, \ldots, a_{r-1}) \notin \mathcal{R}_e(x_{<s}).$$

The source argument is the one fixed true source letter. An origin is therefore a forbidden future set of $r-1$ literals. It always contains a false literal: otherwise the realized word would violate its original relation.

A projection of an origin removes some true literals whose positions have been explicitly recorded. An origin becomes irrelevant when any of its required literals is excluded from a palette. An origin may also be discarded at the chronological retirement of an unrecorded position; this latter loss will be charged to the bad-edge budget.

A.2.1 Pair multiplicity lemma

For any two future literals $v = (i, a)$, $w = (j, c)$ at distinct positions, at most $D = CQ^{r-3}$ complete origins contain both.

Proof. There are at most C original positional edges containing the two positions. In any origin whose future set contains both literals, its source is earlier than both and its source letter is already fixed to the true letter. The two prescribed future letters leave only $r - 3$ future letters unspecified. There are at most Q^{r-3} possibilities. Sum over the at most C edges. □

This is a bound on distinct full origins over the entire execution for this word. It is not a bound on the number of historical projection shapes. The distinction is essential: an origin can appear successively at several ranks, but its complete label is unchanged.

One may define the universe of all complete origins at the outset for purposes of counting, even though the actual algorithm enumerates an origin only when its source is processed. Thus an origin selected just before synchronous palette deletions can be charged to those deletions even if its queue job has not yet been materialized.

A.3 State and invariants

Each position i has a palette S_i, initially Ω. Palettes only shrink, and the true letter is always retained.

A position is **recorded** when its true letter is put in the fingerprint. Its palette then becomes the singleton $\{z_i\}$. A recorded true literal remains in that palette but is inactive in all graphs.

An unprocessed, unrecorded literal in its current palette is active. At the chronological retirement of a position, all its remaining active literals become inactive permanently. Inactivity and palette exclusion are different operations.

For each $2 \le j \le r - 1$, maintain a simple j-uniform hypergraph J_j on active literals. Every physical hyperedge carries one or more complete-origin labels. All labels carried by a physical edge have that edge as their current projection.

Also maintain a finite queue of jobs. A job contains a projected set of literals, the full-origin labels it owns, and a frozen rank bound. Its actual remaining arity may fall below that bound when recorded literals are substituted or excluded literals are noticed. The bound never increases along an origin's history.

We maintain:

I1. Truth preservation. Every palette contains its true letter.

I2. Unique ownership. Each live origin has exactly one current owner: one physical edge or one queued job. Equal physical edges merge their label sets. Origins never branch.

I3. Sound projection. A live projection is obtained from its complete origin only by removing recorded true literals. No unrecorded true value is silently substituted.

I4. No false resurrection. A literal excluded from its palette never returns. An inactive literal never becomes active again.

I5. Degree control. Immediately before every physical insertion, all active degrees in each J_j are strictly less than T_j. At the instant any active literal is removed, its degree in J_j is at most T_j.

I6. Source barriers. Before the chronological retirement of a position, and before the next source is processed, every queue is empty.

All choices—edge order, literal order, witness order, and queue tie breaking—are fixed deterministically from the available state. The full realized word is used only to answer an explicitly posed equality query and to select the heavy or light actual source branch. Replay will eliminate the latter dependence.

A.4 Primitive operations

A.4.1 Excluding a false literal

Delete the literal from its palette and from the active graphs. Every active graph origin requiring it becomes irrelevant. A queued origin requiring it is likewise irrelevant, even if its job discovers this only later. Record the time of the actual palette exclusion for accounting; a later inspection of the job is not a second deletion event.

A negative equality query is one way to exclude a false literal. Unit propagation from a valid forbidden projection is another.

A.4.2 Recording a true literal

For an active literal $v = (i, z_i)$:

1. first snapshot and take ownership of all physical edges incident to v, in every rank, with their origin labels;
2. add (i, z_i) to the fingerprint; exclude all other palette letters at position i; deactivate v, preserving its singleton palette;
3. for every snapshotted rank-two edge $\{v, w\}$, exclude w, before inserting any new physical edge;
4. for every other snapshotted edge B, put $B \setminus \{v\}$, with its origin labels, into a queue with frozen rank $|B| - 1$.

The order can be implemented without ambiguity by first removing all snapshotted physical edges from the graph and giving their labels to their projected jobs or unit consequences. Exclusions can make some of those projected jobs irrelevant before they are later inspected.

Every rank-two neighbor is false: if it were true, an origin of the edge, together with the previously substituted true literals, would make the original forbidden assignment entirely true. Thus I1 is preserved.

Edges contain at most one literal from any position, because they are projections of positional edges. In particular, the other letters excluded at position i do not form second ends of the saved edges incident to v.

A.4.3 Processing a job

If a required letter has been excluded from its palette, discard the job's origins. Otherwise substitute every recorded true literal in it. A letter at a recorded position that does not match the recorded value was already excluded and falls under the first case.

There cannot be zero remaining literals, because then a forbidden origin would be entirely true. With one remaining literal, exclude that false literal. With $j \geq 2$ remaining literals, insert their set into J_j, or merge the origin labels if that physical edge already exists.

A merge does not increase any physical degree and is not a physical insertion.

A.4.4 Degree-triggered queries

Suppose a new physical edge of rank j is inserted. Before insertion every degree is below T_j, so only its ends can reach T_j, and none can exceed it. If an end reaches T_j, choose one such end v and query whether it is true.

A negative response excludes v. A positive response records v. In either case the newly inserted edge disappears immediately. Every other end that had just reached the threshold loses this same new edge and returns below threshold. At that moment other ranks have undergone only removals.

The only remaining operations that can increase degrees are later physical insertions, each handled by the same rule. This proves I5 inductively, once the scheduler below ensures that new insertions are processed one at a time.

A.5 An explicit packet scheduler

A positive query that is triggered in rank j opens its rank-j packet **before** the snapshot and recording operation. Thus the packet includes synchronous exclusion of the other letters at the recorded position and all saved rank-two neighbors.

After recording, drain all pending jobs whose frozen rank is strictly less than j. Each such job has actual arity at most its frozen rank, so any positive query it triggers has rank strictly less than j. That query opens a nested packet of strictly smaller rank. On return, continue draining eligible jobs until none remains. Then close the rank-j packet.

Jobs of frozen rank at least j are deferred while this packet is open. A parent packet can process them later if their frozen rank is below the parent's bound. An external drain with bound $+\infty$ processes all jobs.

This scheduler yields three facts used below:

1. nested positive packets have strictly decreasing ranks;
2. two packets of the same rank have disjoint time intervals;
3. an origin that started as the projection of a saved rank-j edge can never be stranded in the rank-j packet's deferred queue: its initial projection has frozen rank $j-1$, and every subsequent projection along that origin's chain has still smaller frozen rank.

A job that came from an unrelated larger edge can have actual arity below j while retaining frozen rank at least j; deferring it is harmless. The third fact concerns the selected own origins of the packet, whose frozen ranks are already below j.

There is no chronological retirement inside a packet. Chronological retirement happens only after the external drain has finished.

Termination is finite. Only finitely many original forbidden assignments can be installed. Each origin has one chain, and every transfer created by recording strictly reduces its number of required literals. A job is processed once unless it is replaced by such a strict projection; it is otherwise discarded or attached to one physical edge. No origin is duplicated. There are therefore finitely many insertions, transfers, and query events.

A.6 Processing sources

Process sources $i = 0, \dots, N-1$ in order. Queues are empty at entry. The relations whose source is i are known from $x_{<i}$.

If i is already recorded, assign $A_i = \{z_i\}$, enumerate every forbidden future tuple with source letter z_i in the current future palettes, and externally drain all resulting jobs.

If i is not recorded, call a candidate $a \in S_i$ **heavy** if at least T_{s} outgoing positional edges admit a forbidden future tuple in the current future palettes when their first letter is a. Otherwise it is light.

If the true letter is heavy:

- snapshot the current palettes and the full list of forbidden future tuples for the true source letter;
- begin the source accounting interval before any recording or deletion;
- record the source, assign $A_i = \{z_i\}$, and enqueue the snapshotted list;
- externally drain every queue.

If the true letter is light, assign A_i to be all light letters of S_i. This nonempty set contains the true letter. This branch creates no new constraints, no new jobs, and no queries.

Finally, with all queues empty, retire the remaining active literals at position i. Delete all physical edges incident to them, discarding their origins without substituting the unrecorded actual value. Do not change the assigned A_i afterward.

The future positions of any newly installed origin are later than its source. Thus a newly installed origin never contains an already retired position. At a chronological retirement, every earlier queued job has been processed; all still-live origins requiring that position are on its active incident edges and are discarded then. Consequently no later event refers to a retired position. This also proves that future palette changes never alter an already assigned alphabet.

A.7 Physical insertion budget

Let I_j be the number of physical insertions in rank j. Every inserted edge is eventually removed, because all its positions are ultimately retired or recorded.

Charge an insertion to the first removal of one of its active endpoints. Each literal is removed from active status once, and its degree at that event is at most T_j. There are QN literals. Therefore

$$I_j \le T_j QN. \tag{7.1}$$

A physical edge with a given literal set cannot disappear and later be reinserted with that same literal set: its first disappearance removes at least one of those active literals permanently. This is also a direct reason why it is legitimate to speak of the unique insertion event of a surviving physical edge. Alternatively, charging by insertion occurrence gives the same estimate.

A.8 Positive-query budgets: the central charge maps

Let H_j be the number of positive rank-j queries, and let H_{s} be the number of newly recorded heavy sources.

A.8.1 Rank two

At a positive rank-two query, the queried literal has exactly T_2 distinct physical neighbors. All are active and present in their palettes at that instant. Recording removes these T_2 distinct false neighbors before any new insertion.

Each literal can be excluded from its palette at most once. Hence

$$H_2 T_2 \le QN. \tag{8.1}$$

Neighbors may belong to the same position but be different letters; they are still different literals, which is exactly what the QN budget counts.

A.8.2 Rank $j \geq 3$: selection

At a positive rank-j query with true pivot v, its snapshot has exactly T_j incident physical edges. Select one complete-origin label from each. These labels are distinct by unique ownership.

Immediately after substituting v, every selected origin has a projection of rank $j-1$. Such an origin is never selected again by a different rank-j packet, because its single chain can never return to rank j. Thus there are exactly $H_j T_j$ distinct rank-j selected-origin instances over all such packets.

A.8.3 Rank $j \geq 3$: exhaustive exit alternatives

For each selected origin, at the closure of its packet exactly one of the following accounting alternatives applies:

- it has lost a required palette letter during this packet;
- it is carried by an active physical edge of rank $l < j$.

Here is why this list is exhaustive. No chronological retirement occurs within the packet. The origin started alive and its actual projection has rank $j-1$. Every subsequent job on its chain has frozen rank below j, so cannot remain in the deferred queue when the packet closes. If a recorded end is substituted, the rank decreases further. A zero-end origin is impossible by soundness. A one-end origin causes a palette exclusion. If it is neither irrelevant nor still queued, unique ownership places it on an active physical edge, necessarily of smaller rank.

A.8.4 Charging actual palette losses

For the first alternative, charge the selected origin to its first required palette exclusion during the packet. That excluded literal w and the packet pivot v both belong to the origin's full future assignment. They are distinct: the true pivot is recorded and retained, not excluded.

For one fixed packet and one fixed exclusion event w, the pair lemma bounds these selected origins by D. Packets of the same rank are time-disjoint, so an actual exclusion event belongs to at most one rank-j packet. There are at most QN exclusion events overall. The total loss charge is therefore at most DQN.

This includes exclusions at packet opening. An origin whose queued projection is inspected later is charged to the earlier actual exclusion event within the same packet. Nothing is charged to a merely administrative late inspection.

A.8.5 Charging surviving lower edges

For the second alternative, charge the selected origin to the unique insertion event of its final active physical edge B of rank $l < j$.

Fix any pair of literals in B. Every complete origin charged to this edge contains that pair. Across all rank-j packets, each complete origin is selected at most once at rank j, and at most D complete origins contain the pair. The physical insertion consequently pays for at most D selected origins, even if some origins merge into an edge created before their packets began.

This is the precise resolution of the "old lower edge receives new projections" concern. The budget counts its full-origin labels over time, not its creation time relative to the paying packet.

Combining both charge classes gives

$$H_j T_j \leq D\left(QN + \sum_{l=2}^{j-1} I_l\right), \qquad 3 \leq j \leq r-1. \tag{8.2}$$

A.9 Heavy-source budget

At a newly recorded heavy source choose T_{s} different outgoing positional edges witnessing heaviness, and one forbidden future tuple from each, in the palettes snapshotted before recording.

Their future position sets are pairwise disjoint. These tuples are alive at the start of the source accounting interval. At the end of the external drain, a selected origin either has lost a palette letter during this interval or is on an active physical edge. There are no queued jobs, no chronological retirements during the interval, and no completely true forbidden tuple.

An actual excluded literal belongs to at most one selected future tuple for this source, by future-position disjointness. Source intervals are successive and disjoint. Thus palette-loss charges total at most QN.

For surviving-edge charges, any fixed pair of the active edge belongs to every paying complete origin; a source is processed only once and one origin was chosen for each selected outgoing edge. The pair lemma bounds charges to a physical insertion by D, over all source intervals.

Therefore

$$H_{\mathrm{s}}T_{\mathrm{s}} \le QN + D\sum_{j=2}^{r-1} I_j. \tag{9.1}$$

It is legitimate for a deletion or insertion to participate in the separate upper bounds for different ranks and for heavy sources. These are different estimates for different record counts; each estimate has its own valid charge map. The proof never asserts a single shared global budget across all these independently bounded classes.

A.10 Fingerprint size

A position is recorded only once. Every recording is either a new heavy-source recording or a positive query recording. Thus

$$|F| = H_{\mathrm{s}} + \sum_{j=2}^{r-1} H_j.$$

Using (7.1), (8.1), (8.2), and (9.1) gives

$$H_{\mathrm{s}} \le QN\frac{D\sum_{j=2}^{r-1} T_j + 1}{T_{\mathrm{s}}},$$

$$H_2 \le QN/T_2,$$

$$H_j \le DQN\frac{1+\sum_{l=2}^{j-1} T_l}{T_j} \quad (j \ge 3).$$

Their sum is exactly θN, with the stated constant. No hidden dependence on N, the relations, or the realized word appears.

A.11 Bad-edge budget

Call an edge bad for this word if its assigned full Cartesian product fails to be contained in the relation frozen at its actual source prefix.

A.11.1 Light sources

For a light source, each of its at most Q assigned source candidates has fewer than T_{s} outgoing edges with a forbidden future tuple in the then-current future palettes. Later future alphabets are subsets of those palettes. Consequently the source has fewer than QT_{s} bad outgoing edges.

The total light-source contribution is at most $QT_{\mathrm{s}}N$.

A.11.2 Recorded sources

For a bad edge with recorded source, choose one forbidden tuple from the final assigned alphabets. Its source letter is the actual recorded letter. Its future letters were all present when its source was processed, because palettes shrink and assigned alphabets are never enlarged after retirement. Hence this full origin was enumerated or was already rendered irrelevant only by a palette event that would also exclude a final alphabet letter.

The latter cannot happen: every chosen future letter belongs to its assigned final alphabet, so none was ever excluded from its palette. Recording one of its future positions can only substitute the matching true letter; recording a different letter would exclude the chosen one, again contradicting final membership.

Follow the origin's unique chain. It cannot end in a unit exclusion or an empty projection, and it cannot disappear through a false-letter exclusion. At the end of the complete execution no active literal remains. Because every chronological retirement has an empty queue, the origin must first be discarded on an active physical edge at a chronological retirement.

Charge this bad positional edge to that physical edge's insertion event, using the selected complete origin as the invoice. Different bad positional edges have different complete origins. Every origin charged to a fixed physical edge contains any chosen pair of its literals. The pair lemma therefore bounds its charges by D.

The recorded-source contribution is at most $D\sum_j I_j$. Together with (7.1),

$$E_{\mathrm{bad}} \le QT_{\mathrm{s}}N + D\sum_{j=2}^{r-1} I_j \le Q\left(T_{\mathrm{s}} + D\sum_{j=2}^{r-1} T_j\right)N.$$

This bound explicitly pays for chronological deletion without substituting an unrecorded actual value. Omitting that payment would leave a real gap; it is included in the bound here.

A.12 Exact replay from the final fingerprint

Fix a realized pair $(x_{<i}, F(x))$. We show that the entire execution through assignment of A_i can be reconstructed from this pair.

At any equality query on an active literal (k, a), the answer is positive if and only if $(k, a) \in F(x)$:

- every positive answer immediately records the pair;
- every recorded pair is true;
- a true literal cannot be excluded or change its value later.

The fingerprint is used only to answer the queries actually produced by the deterministic algorithm. It is not used to insert all future recorded letters into the state in advance. Absence of an unqueried pair says nothing by itself about that letter.

Process earlier sources by induction. Their relations are known because their source prefixes are restrictions of $x_{<i}$. Each queue action and literal query is then determined by the reconstructed state and the fingerprint oracle.

The only remaining branch is a source $s \le i$ that has not already been recorded when its turn arrives. At that point all queues are empty.

- If $s \notin \operatorname{dom} F$, its actual source letter cannot be heavy, because that would record it. Its actual letter is light. The algorithm computes A_s as all light letters from the known state and creates no events.
- If $s \in \operatorname{dom} F$, its true letter is given by its fingerprint pair. If that letter were light, the light branch would create no events; the source would immediately retire, after which it could never be recorded. This contradicts membership in the final fingerprint. Therefore the actual letter is heavy and the heavy branch is determined.

An already recorded source uses its known recorded value. This completes the induction. The algorithm determines A_i from $(x_{<i}, F)$ exactly as claimed.

Consequently, for any probability distribution on $\mathcal{W}$, the assigned A_i contains the entire conditional support

$$\operatorname{supp}(Z_i \mid x_{<i}, F),$$

on every conditioning atom of positive probability. This is stronger than merely containing the one actual letter on one execution.

A.13 Fingerprint entropy

There are at most

$$\sum_{a=0}^{\lfloor \theta N \rfloor} \binom{N}{a} Q^a$$

possible fingerprints. Since $\theta \le 1/2$,

$$\sum_{a \le \lfloor \theta N \rfloor} \binom{N}{a} \le \exp\{N h(\theta)\}, \qquad Q^a \le Q^{\theta N}.$$

Thus

$$H(F) \le \log |\operatorname{range} F| \le N[h(\theta) + \theta \log Q].$$

The map F is deterministic, but this count gives the same bound for every distribution on $\mathcal{W}$. The proof of the finite theorem is complete.